

AI Summaries Overrepresent Fake Reviews: Evidence from Amazon

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Abstract

AI summarization has been widely deployed to distill information from large volumes of consumer reviews. In this research, we argue that AI summarization produces an unintended consequence: the overrepresentation of fake reviews. Our key insights are that (i) AI summarization is designed to extract common themes and (ii) fake reviews are more linguistically similar to one another, making them more likely to share common themes. To provide empirical support, we analyze AI summaries on Amazon. With both review-level and product-level analyses, we consistently find that AI summaries overrepresent fake reviews. We further examine the impact of the overrepresentation of fake reviews on market outcomes, finding that review manipulators receive more positive AI summaries than other sellers, even after controlling for average ratings and other observable characteristics. Furthermore, we find that review manipulators experience a significantly greater improvement in sales rank following the introduction of AI summaries relative to their competitors.

Keywords: AI Summary, AI Bias, Fake Reviews, Digital Platforms, Misinformation

1 Introduction

With the development of AI technologies, digital platforms (e.g., Amazon, Walmart, Best Buy, Yelp, Apple App Store, and Google Play Store) have recently launched AI summarization of reviews. According to Vaughn Schermerhorn, the director of community shopping at Amazon, AI summarization is intended to help consumers grasp common themes more easily from the vast volume of reviews on the platform.¹ AI summarization typically involves two components: (i) a natural language processing model that extracts common topics from reviews, and (ii) sentiment analysis for each topic. A short paragraph is then composed based on the extracted topics and their average sentiments. Crucially, not all reviews are treated equally in this process. Specifically, reviews that mention more extracted topics contribute more to AI summaries, while reviews that do not mention common keywords extracted by summarization algorithms tend to be ignored.

Although this process of generating AI summaries seems reasonable, consumer reviews often include fake reviews (Mayzlin et al., 2014). Major media outlets (e.g., the Wall Street Journal and the Associated Press) have warned that fake reviews are widespread,^{2,3} and that they can even fool experienced shoppers.⁴ This concern has prompted regulatory and legislative action, including a recent rule issued by the Federal Trade Commission (FTC) in the US⁵ and legislation enacted by the UK Parliament,⁶ to combat fake reviews.

Building on this institutional background, we argue that AI summaries may unintentionally overrepresent fake reviews. This occurs because fake reviews tend to emphasize generic product merits and thus contain more common themes; simultaneously, AI summarization algorithms are explicitly designed to extract these prevalent patterns. Consequently, AI summaries may systematically overrepresent fake reviews. Furthermore, because the tone of fake reviews is typically more positive than that of authentic ones even after controlling for star ratings, this overrepresentation renders the summaries of manipulated products overly positive, potentially boosting the sales of review manipulators.

In this paper, we define the overrepresentation of fake reviews as a condition where, on average, a fake review mentions more extracted common themes than an authentic review. Building on this definition and our arguments above, we hypothesize that AI summaries overrepresent fake reviews, leading to two distinct market effects: (i) review manipulators receive more positive AI

summaries than other sellers, and (ii) review manipulators experience a greater increase in sales following the introduction of AI summaries relative to their competitors.

To test our hypotheses regarding the overrepresentation of fake reviews in AI summaries and its subsequent market effects, we assemble a dataset from multiple sources: Amazon, Keepa, and RateBud. We use Amazon-sold products as a proxy for items less likely to contain fake reviews, as Amazon rarely engages in fraudulent review manipulation (He et al., 2022b). Conversely, products with low review-credibility grades on RateBud⁷ serve as a proxy for products with a higher prevalence of fake reviews. Keepa provides sales rank data, which allows us to test whether AI summaries potentially enhance review manipulators' sales. We further supplement our data by leveraging fake reviews identified in prior literature (He et al., 2022a; Feldman et al., 2025).

It is challenging to empirically test the overrepresentation of fake reviews, because we do not directly observe either which reviews are fake or how many extracted common themes each review mentions. To tackle this empirical challenge, we adopt two complementary approaches. First, we conduct a review-level analysis. We leverage the fake review tags in the dataset of He et al. (2022a) and Feldman et al. (2025), and run modern natural language processing algorithms to determine whether one review mentions the extracted keywords. Across specifications, we consistently find that fake reviews mention more extracted keywords than authentic reviews, suggesting that AI summaries overrepresent fake reviews. Second, we conduct a product-level analysis using observables displayed on the user interface of Amazon. We find that products with a higher share of fake reviews also have a greater average number of mentions of the extracted keywords per review. This provides further empirical support for the overrepresentation of fake reviews using the output of Amazon's deployed AI summarization algorithms on the full set of consumer reviews, rather than only on the subsample of reviews that can be classified as fake or authentic with sufficient confidence.

Next, we investigate the effects of overrepresentation of fake reviews on market outcomes. First, we examine whether products with a higher share of fake reviews receive systematically more positive AI summaries. Using OpenAI's GPT-4.1 and a fine-tuned BERT model specialized for review sentiment, we analyze the sentiment of AI summaries. We find evidence that products with a greater percentage of fake reviews receive more positive AI summaries, both in terms of the sentiment labels assigned to the extracted keywords (which correspond to common themes) and

the sentiment of AI summary paragraphs. This suggests that the overrepresentation of fake reviews fundamentally distorts product quality information in the market. Furthermore, we test whether this distortion in product quality information translates into distorted market outcomes. We find that following the introduction of AI summarization, the sales ranks of Amazon-sold products (which are significantly less susceptible to manipulation) drop significantly, while products with low review-credibility grades on RateBud see an average improvement in their sales ranks relative to their competitors.

This paper contributes to three primary streams of literature. First, we document a new form of AI bias, arising as an unintended consequence of the interaction between the mechanistic nature of keyword extraction in AI summarization algorithms and the specific textual features of fake reviews. Second, we contribute to the literature on fake reviews by demonstrating that AI summaries can heighten the salience of fraudulent information, thereby improving the sales ranks of manipulated products, a shift that potentially harms consumer welfare. Finally, we add to the literature on misinformation transmission by showing that AI can function as an amplifier of misinformation, specifically in the context of deceptive fake product reviews.

The findings of this paper hold significant implications for policymakers. If platforms continue to deploy AI summarization of reviews without addressing the systematic bias documented here, the resulting market distortions could lead to substantial welfare loss for consumers. Furthermore, our research is critically relevant to platform managers focused on long-term customer trust; by inadvertently directing consumers toward sub-standard products, current AI summarization algorithms may fuel customer dissatisfaction and erode brand equity. Ultimately, we hope this work prompts AI developers to refine algorithms so they can more robustly summarize information in environments contaminated by misinformation.

The remainder of the paper is organized as follows. Section 2 reviews the literature and elaborates on our contributions to the literature. Section 3 introduces the institutional background. Section 4 develops a theoretical framework that defines the overrepresentation of fake reviews and proposes hypotheses. Section 5 introduces data sources. Section 6 reports the empirical evidence that AI summaries overrepresent fake reviews. Section 7 presents empirical results that review manipulators receive more positive AI summaries and that the introduction of AI summarization increases sales of review manipulators. Section 8 concludes with the implications of the findings.

2 Literature Review

Our work intersects three streams of literature: AI bias, fake reviews, and the spread of misinformation. The literature on AI bias and algorithmic bias has attracted attention from scholars in various fields. [Cowgill and Tucker \(2019\)](#) provide a comprehensive literature review of algorithmic bias. They distinguish algorithmic bias caused by biased objectives from algorithmic bias caused by biased predictions, highlighting the particular challenges in dealing with the former. Solving algorithmic bias caused by biased objectives is a governance problem, which involves leadership, ethics, and oversight, rather than pure technical tweaks. A classic example is the filter bubble. Researchers ([Levy, 2021](#); [Nyhan et al., 2023](#); [González-Bailón et al., 2023](#)) have confirmed that recommendation algorithms expose users to more like-minded content that is consistent with users' own political beliefs. Other examples include [Obermeyer and Mullainathan \(2019\)](#), in which they find racial disparities in a commercial health risk prediction algorithm that is mistakenly optimized for healthcare cost rather than actual health. For another example, [Lambrecht and Tucker \(2019\)](#) find that STEM job information is more likely to be allocated to a male audience by recommendation algorithms. Our research contributes by documenting a new example of AI bias induced by biased objectives. We show that AI summarization favors fake reviews because it is unintentionally designed to seek textual features typically possessed by fake reviews.

We also contribute to the literature on fake reviews. Online reviews have been shown to influence the behaviors of both consumers ([Chevalier and Mayzlin, 2006](#); [Zhu and Zhang, 2010](#); [Sun, 2012](#); [Lu et al., 2022](#); [Wang et al., 2025](#)) and sellers ([Farronato and Zervas, 2022](#)), and consequently influence consumer welfare ([Wu et al., 2015](#)). Therefore, fake reviews have raised widespread concerns. [Mayzlin et al. \(2014\)](#) are among the first to show convincing evidence of the existence of fake reviews, and [Luca and Zervas \(2016\)](#) investigate the economic incentives underlying review manipulation. [He et al. \(2022b\)](#) contribute significantly to the literature by directly observing the market for fake reviews, and therefore more confidently identifying fake reviews. Not only do they provide details of the market for fake reviews, they also shed light on whether review manipulation hurts consumer welfare ([Dellarocas, 2006](#); [Mayzlin, 2006](#)). [Gandhi et al. \(2024\)](#) further study the welfare implications of fake reviews. Our paper contributes to this stream of literature by suggesting that the negative welfare impact of fake reviews might be reinforced by AI summaries

in the AI era.

Our paper is related to the literature on the transmission of misinformation. [Vosoughi et al. \(2018\)](#) provide strong empirical evidence that misinformation diffuses significantly faster, farther, deeper, and more than authentic information. Researchers are interested in the reasons. In the theoretical literature, [Acemoglu et al. \(2010\)](#), [Acemoglu et al. \(2013\)](#) and [Mostagir et al. \(2022\)](#) identify social network structures that are susceptible to information manipulation. The existing empirical studies also find evidence that robots ([Shao et al., 2018](#)), the tendency of human beings to engage more with misinformation ([Vosoughi et al., 2018](#)), the failure of human beings to consider accuracy of information ([Pennycook et al., 2021](#)), overconfidence in the ability to spot misinformation ([Lyons et al., 2021](#)), political polarization ([Zhu and Pechmann, 2024](#)), and echo chambers ([Vicario et al., 2016](#); [Törnberg, 2018](#)) can facilitate the spread of misinformation. Our paper contributes to this stream of literature by showing that AI can also amplify the voice of false information, which is fake reviews in our context on digital platforms.

3 Institutional Background

3.1 AI Summary of Reviews

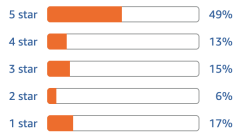
With the development of AI technologies, most digital platforms, including Amazon, Walmart, Best Buy, Yelp, Apple App Store and Google Play Store, have adopted AI summarization of reviews. AI summaries of reviews are machine-produced and language-model-based condensations of user-generated reviews that aim to mitigate information overload and support consumer decision-making on digital platforms.

AI summaries are generally displayed at a prominent location above all user-generated reviews. For example, [Figure 1](#) shows how AI summaries are displayed on Amazon and Best Buy. AI summaries typically include two user-facing parts, a list of keywords with corresponding sentiments, and a summary paragraph. Some platforms have both (e.g., Amazon and Best Buy), some only have keywords with sentiments (e.g., Yelp), and others only have summary paragraphs (e.g., Apple App Store). When both keywords with sentiments and summary paragraphs are present, summary paragraphs typically connect keywords into a human-understandable summary paragraph so that the two user-facing parts are consistent with each other.

Figure 1: User Interfaces of AI Summaries**(a) Amazon⁸****Customer reviews**

★★★★☆ 3.7 out of 5

174 global ratings



How customer reviews and ratings work ▾

Review this product

Share your thoughts with other customers

Write a customer review



Timekettle M3 Languanoe

Customers say

Customers give positive feedback about the translation earbuds' quality and communication ability, with one customer noting how it helps communicate with locals with confidence. However, the translation performance receives mixed reviews, with some finding it accurate while others report slow Thai to English translation. Moreover, the functionality, ease of use, and value for money are also mixed aspects, with some finding it worth the price while others consider it a waste of money. Additionally, customers report a delay of 2 to 5 seconds during translation.

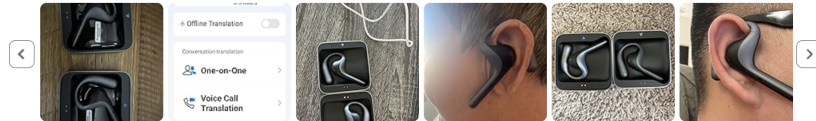
AI Generated from the text of customer reviews

Select to learn more

✓ Quality | ✓ Communication ability | Translation quality | Functionality | Ease of use |
Value for money | Language support | ⌚ Lag time

Summary Paragraph**Keywords and Sentiments****Reviews with images**

See all photos >

**Top reviews from the United States**

Amazon Customer

★★★★★ A truly life changing translator

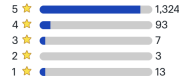
Reviewed in the United States on September 18, 2025

Verified Purchase

(b) Best Buy⁹**Reviews**

★ 4.9

1,440 reviews



✓ 98% would recommend to a friend

Top Mentions

- Overall Performance (212)
- Portability (74)
- Battery Life (55)
- Speed (50)
- Screen Quality (40)
- Refresh Rate (4)

Keywords and Sentiments**Customers are saying**

Customers admire the 11-inch iPad Air M3's overall performance, particularly praising its speed and ability to handle demanding tasks. Its portability and excellent screen quality are also frequently highlighted, with many users appreciating the lightweight design and vibrant display. Positive feedback also includes the improved battery life and the aesthetically pleasing design. While some users mentioned the refresh rate, the majority found it acceptable.

This summary was generated by AI based on customer reviews.

Customer Images

The vast majority of our reviews come from verified purchases. Reviews from customers may include My Best Buy members, employees, and Tech Insider Network members (as tagged). Select reviewers may receive discounted products, promotional considerations or entries into drawings for honest, helpful reviews.

★★★★★

iPad Air m3 11"

Incentivized | Verified Purchaser | Owned for 1 month

The new iPad Air works great for on the go, light weight, mini workstation. If you are in school/work, it is a great alternative to a laptop with the powerful m3 chip. It

Summary Paragraph

In this paper, we focus on Amazon. More than half (56%) of consumers in the US start their shopping searches on Amazon.¹⁰ Major media noticed AI summaries of reviews on Amazon in the middle of June 2023.^{11,12} Amazon formally announced AI summarization of reviews on August 14, 2023.¹ Over time, AI summarization was gradually rolled out to a growing number of products (Su et al., 2025). Regarding the business goal of AI summarization, Vaughn Schermerhorn, the director of community shopping at Amazon, claimed that they “want to make it easier for customers to understand the common themes across reviews.” Moreover, he mentioned that “(AI summarization) provides a short paragraph right on the product detail page that highlights the product features and customer sentiment frequently mentioned across written reviews to help

customers determine at a glance whether a product is right for them.”

A document from the internal AI team of Amazon describes the basic features of the AI summarization algorithm.¹³ According to Amazon engineers, the AI summarization algorithm “assigns sentiment analysis and keyword extraction to traditional ML while using optimized SLMs (small language models) for complex text generation tasks”. In other words, Amazon first feeds user-generated reviews into specialized traditional machine learning models to extract keywords that are most commonly mentioned across reviews. Amazon only uses reviews that mention this set of the most popular keywords for sentiment analysis. The sentiment analysis evaluates the tone of each review that mentions the extracted keywords (by coding it as positive or negative), and aggregates them to its overall sentiment as either positive, neutral, or negative. Finally, Amazon uses small language models to generate summary paragraphs.¹⁴ It is important to highlight one key difference between AI summarization and traditional methods, such as average ratings. Instead of extracting information equally from all reviews, the AI summarization algorithm only considers sentiments of reviews that mention the extracted keywords, while ignoring the remaining reviews. Apple deploys a similar algorithm in their App Store.¹⁵

To our knowledge, the existing research on AI summarization is still work-in-progress. [Wang et al. \(2025\)](#) and [Wang and Wang \(2025\)](#) find that the introduction of AI summarization increases purchase rates and shifts users from reading more reviews to exploring more listings. [Su et al. \(2025\)](#) study the impact of AI summarization on the content richness and the number of following user-generated reviews. To the best of our knowledge, no existing research has studied whether misinformation is overrepresented by AI summaries.

3.2 Fake Reviews

The market for fake reviews works as follows. Third-party sellers on Amazon who attempt to manipulate reviews (i.e., review manipulators) create private groups on social media platforms such as Facebook, Twitter, and Telegram to recruit and communicate with fake reviewers.¹⁶ These review manipulators offer consumers deals to write 5-star reviews for them in exchange for a full refund and, sometimes, an additional payment ([Huang et al., 2023](#)). Due to the high costs of buying fake reviews and the large number of third-party sellers in the same market, there are very few

negative fake reviews attacking competitors on Amazon (He et al., 2022b). Also related to the high price of fake reviews, according to Saoud Khalifah, the founder of Fakespot, the main buyers of fake reviews are sellers of low-price products (around \$15 - \$40).¹⁶ Prior research also finds that Amazon does not engage in this type of fake review practice for products that are directly sold by Amazon (He et al., 2022b).

Some research also studies the textual features of fake reviews. Ott et al. (2011) find that fake reviews contain more superlatives, have more positive and fewer negative emotion terms, and include less concrete language than authentic reviews. Li et al. (2014) show that fake reviews exhibit excessively strong sentiments. Luca and Zervas (2016) also have similar findings in the reviews filtered and then removed by Yelp. Moreover, previous research has noticed that many fake reviews are highly similar to each other (e.g., Jindal and Liu, 2008; Mukherjee et al., 2012; Rayana and Akoglu, 2015; Wang et al., 2023; Gupta et al., 2024), and that fake reviewers tend to work collaboratively in groups (Rathore et al., 2021). Additionally, He et al. (2022b) notice that fake reviews are longer than authentic reviews on Amazon.

Textual features of fake reviews are consistent with how the market of fake reviews works. When fake reviewers have to write favorable comments for an item that they are not very familiar with, they tend to include common and general merits, making fake reviews resemble each other. In contrast, when consumers have idiosyncratic good personal experiences with the products, they tend to include more heterogeneous details. On platforms with strict anti-fake-review policies, such as Amazon, sellers need to reimburse fake reviewers for purchasing the products in the vast majority of cases, which raises manipulation costs; consequently, paid fake reviewers are asked to write longer and more positive reviews.

4 Theoretical Framework

In this section, we first define the overrepresentation of fake reviews. Then, we hypothesize that AI summaries overrepresent fake reviews and propose two effects of overrepresentation on product quality information and sales outcomes.

4.1 Definition of Overrepresentation

AI summaries of reviews represent consumer reviews that mention the extracted keywords. Reviews that mention more of these extracted keywords appear more in AI summaries, receive greater representation in AI summaries, and are therefore more likely to shape the content and sentiment of AI summaries. We say that fake reviews are overrepresented if, on average, a fake review mentions more keywords extracted by AI summarization algorithms than an authentic review.

More formally, let W_{ij} denote the number of extracted keywords that review i of product j mentions, and let $F_{ij} \in \{0, 1\}$ indicate whether the review is fake. $F_{ij} = 1$ if review i of product j is fake and $F_{ij} = 0$ if review i of product j is authentic.

Definition 1 (Overrepresentation of Fake Reviews). *We define the overrepresentation of fake reviews as a condition in which, on average, a fake review mentions more keywords extracted by AI summarization algorithms than an authentic review, i.e., $\mathbb{E}[W_{ij} \mid F_{ij} = 1] > \mathbb{E}[W_{ij} \mid F_{ij} = 0]$.*

This definition is consistent with the literature on overrepresentation. In that literature, overrepresentation typically occurs when individuals with certain characteristics appear in a salient representation more frequently than others, thereby making the representation biased. For example, prior studies examine overrepresentation in settings such as children’s books (Adukia et al., 2023), juries (Anwar et al., 2022), and public-sector employment (Gomes and Kuehn, 2025). Our definition follows the same logic: fake reviews are overrepresented if they appear in AI summaries more frequently than authentic reviews.

4.2 Intuition for Overrepresentation

As explained in Section 3.1, AI summarization is designed to summarize common themes across different consumer reviews instead of idiosyncratic opinions.¹ As introduced in Section 3.2, there is a consensus in the literature that compared to authentic reviews, fake reviews are more similar to each other (Jindal and Liu, 2008; Mukherjee et al., 2012; Rayana and Akoglu, 2015; Wang et al., 2023; Gupta et al., 2024). Ott et al. (2011) notice that fake reviews include less concrete language and are less specific than authentic reviews. Additionally, on Amazon, they are also longer (He

et al., 2022b). All these findings suggest that fake reviews are likely to repeat generic and recurring themes.

Therefore, it is reasonable to hypothesize that AI summaries overrepresent fake reviews. We make Hypothesis 1, and empirically test whether the textual features of fake reviews translate into more common themes that can be detected by AI summarization algorithms.

Hypothesis 1 (Overrepresentation of Fake Reviews). *AI summaries overrepresent fake reviews, i.e., $\mathbb{E}[W_{ij} \mid F_{ij} = 1] > \mathbb{E}[W_{ij} \mid F_{ij} = 0]$.*

4.3 Effect 1: Distortion in Product Quality Information

We discuss the negative effects of the overrepresentation of fake reviews. On Amazon, fake reviews are almost always positive for review manipulators themselves, because purchasing fake reviews is costly (He et al., 2022b). According to the institutional background introduced in Section 3.2, fake reviews are deliberately asked to be positive. The overrepresentation of the overly positive portion among all reviews of review manipulators can make AI summaries of review manipulators more positive. This leads to Hypothesis 2.

Hypothesis 2 (Distortion in Product Quality Information). *Compared with other products, products with a higher share of fake reviews receive more positive sentiments corresponding to extracted keywords and more positive AI summary paragraphs.*

4.4 Effect 2: Distortion in Market Outcomes

Researchers have long been aware that online reviews strongly influence sales (Chevalier and Mayzlin, 2006; Zhu and Zhang, 2010; Sun, 2012; Lu et al., 2022; Wang et al., 2025; Wang and Wang, 2025). AI summaries of reviews are positioned in a more prominent location above all user-generated reviews. Hence, AI summaries of reviews should have a stronger influence on sales and consumer decision-making. If Hypothesis 2 is true, we should observe that after the introduction of AI summarization, review manipulators experience greater sales compared with non-manipulators. This leads to Hypothesis 3. Hypothesis 3 is a market distortion in which cheaters benefit in sales outcomes while truth-tellers suffer. Such distortion may direct consumers to purchase sub-optimal products and encourage more sellers to manipulate reviews.

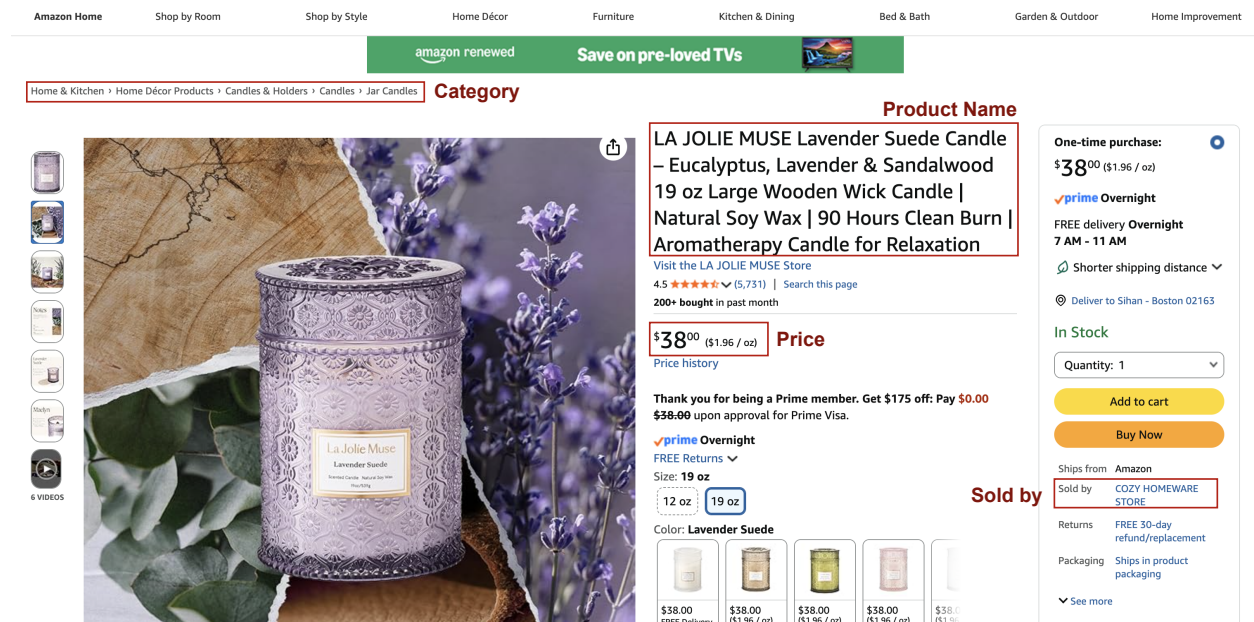
Hypothesis 3 (Distortion in Market Outcomes). *After the introduction of AI summarization, compared with other products, products with a higher share of fake reviews experience greater improvement in sales.*

5 Data

This paper relies on four data sources. The first is Amazon. From Amazon, we download (1) AI summary paragraphs, (2) extracted keywords with corresponding sentiment labels, (3) the number of mentions of each keyword (i.e., the number of reviews that mention each keyword), (4) the total number of reviews, (5) distribution of ratings (shares of 1-star, 2-star, 3-star, 4-star, and 5-star reviews), (6) average ratings, (7) product names, (8) prices of products,¹⁷ (9) categories of products,¹⁸ and (10) whether the product is sold by Amazon. They are all displayed on the Amazon interface. (1) - (6) can be easily observed in Figure 1, while (7) - (10) are highlighted on the Amazon interface shown in Figure 2. When collecting data, we identify the most popular search terms in the US and around the world from reports by commercial consulting firms, including Exploding Topics¹⁹ and Glimpse.²⁰ Then, we search these search terms on Amazon, and record products on the first seven pages of the results.²¹ We collected 16,851 products for 159 search terms in total on September 29, 2025. We report the summary statistics in Table 1. The number of reviews varies a lot across different products. The mean is much larger than the median, suggesting that the number of reviews follows a strongly right-skewed (positively skewed) distribution. A small share of products have many reviews. The price also follows a strongly right-skewed distribution. In contrast, the average ratings have a very small variation. Its distribution is concentrated in the range of 4.3 to 4.7. In the data we collected, around 30% of the products were sold by Amazon. The missing average ratings in the data are mainly due to no reviews, and the missing Amazon-Sold product indicators are mainly because of no items in stock when we collected the data.

The second data source is a dataset compiled by Brett Hollenbeck from various studies, publicly available on his GitHub page.²³ It contains products classified as fake review buyers by He et al. (2022a) and fake reviewers classified by Feldman et al. (2025). We use this dataset to analyze textual features of fake reviews. The product-level summary statistics are reported in Table 1, while the review-level characteristics are reported in Panel A of Table 3. In Table 1, we notice that the number of reviews is much lower than in data source 1. We have confirmed that this dataset includes a

Figure 2: Locations of Product Categories, Product Names, Product Prices, and Amazon-Sold Status on the Amazon Interface²²



subset of all reviews that can be classified as fake or authentic with enough confidence. We classify a review as a fake review if it is (i) a 5-star review, (ii) written for a review manipulating seller classified by He et al. (2022a), and (iii) written by a fake reviewer classified by Feldman et al. (2025). We augment the dataset by collecting lists of keywords of the products in it from Amazon on April 23, 2026.

The third data source is Keepa.²⁴ From Keepa, we obtain the sales rank history of the products in data source 1. Keepa provides the sales rank history of 11,523 products (around 70% of the products in data source 1). Each product is ranked within the first-level category shown in Figure 2. Table 1 reports summary statistics of sales ranks for the full sample and for the top categories. As can be easily noticed, the distribution of sales ranks is very right skewed (positively skewed). This is consistent with the well-known long-tail pattern in online markets (Brynjolfsson et al., 2011). Following the classic paper on online reviews Chevalier and Mayzlin (2006), we use the logarithmic transformation of sales ranks as the dependent variable in our analysis. The history of featured offers is also collected from Keepa to check whether Amazon-sold products in our data source 1 (i.e., Amazon) are mostly sold by Amazon in history and to run robustness checks.

The fourth data source is RateBud.⁷ We collect from RateBud a list of suspicious review manipulators among products in data source 1. RateBud publishes the grades of almost all the products

in data source 1. RateBud comprehensively grades the overall credibility of reviews of products on Amazon according to product consistency (consistency between reviews and product), seller reputation (seller credibility with brand recognition), review distribution (whether distribution of ratings is natural), reviewer credibility (trustworthiness of reviewers), review velocity (the rate and timing of reviews over the product’s history) and review content quality (quality and authenticity of review text). In summary statistics in Table 2, we find that most of the products in data source 1 are of grade A (Excellent) or grade B (Very Good).

Table 1: Summary Statistics for Data Sources 1 (Amazon), 2 (Fake Reviews), and 3 (Keepa)

	Mean	STD	25%	50%	75%	Observations
Data Source 1: Amazon						
# Reviews	6,815.17	24,534.75	125	859	4,242	16,851
Average Rating	4.431	0.334	4.3	4.5	4.6	16,457
Price	175.65	811.48	20.68	42.50	139.99	16,851
Amazon-Sold Product	30.08%					16,812
Data Source 2: Dataset Compiled from He et al. (2022a) and Feldman et al. (2025)						
Full Sample						
# Reviews	112.64	184.41	17	46	127	3,389
Percentage of Fake Reviews	17.93%	27.23%	0	0	32.72%	3,389
Among Products with Both Fake And Authentic Reviews						
# Reviews	109.89	188.78	18	46	116	1,459
Percentage of Fake Reviews	39.26%	26.24%	17.46%	36.36%	60.00%	1,459
Data Source 3: Keepa						
Sales Rank	27,613.98	116,710.55	920	4,257	18,063	38,117,057
- Health & Household	20,079.66	50,231.23	1,686	5,648	17,698	8,685,872
- Electronics	23,376.79	63,338.61	1,026	3,777	16,475	3,455,223
- Home & Kitchen	79,160.81	260,624.62	3,120	13,548	52,738	3,297,919
- Beauty & Personal Care	15,484.78	48,045.74	679	2,925	11,108	3,135,748
Log(Sales Rank)	8.26	2.18	6.82	8.36	9.80	38,117,057

Note: This table presents the summary statistics of data sources 1, 2, and 3. The summary statistics for data sources 1 and 2 are at the product level, while those for data source 3 are based on panel data with both product and time dimensions. In the sample we collected from data source 1, 14,479 products have AI summaries. According to Amazon, AI summaries are shown when “shared mentions” are abundant enough.²⁵ The missing average ratings (394 products) in the data are mainly due to no reviews, and the missing Amazon-Sold product indicators (39 products) are mainly because of no items in stock when we collected the data.

Table 2: Summary Statistics for Data Source 4 (RateBud)

Grade	A	B	C	D	E	F	Total
Meaning	Excellent	Very Good	Good	Fair	Poor	Caution	
Count	9,998	4,777	1,306	276	0	287	16,644
Percentage	60.07%	28.70%	7.85%	1.66%	0%	1.72%	100%

6 Overrepresentation of Fake Reviews

In this section, we test Hypothesis 1 by studying whether AI summaries overrepresent fake reviews. Meanwhile, we also show empirical evidence that is consistent with the intuition for the overrepresentation of fake reviews.

6.1 Review-Level Analysis

6.1.1 Mechanism-Motivating Differences in Textual Features of Reviews

We first describe the basic textual features of reviews including the length, sentiment, and textual similarities. We classify a review as a fake review ($F_{ij} = 1$) if it is (i) a 5-star review, (ii) written for a review manipulating seller classified by He et al. (2022a), and (iii) written by a fake reviewer classified by Feldman et al. (2025). We select products that have both fake reviews and authentic reviews. To compare the length, we simply count the number of characters and words. To compare the sentiments of review texts, we use OpenAI GPT-4.1,²⁶ and set the temperature (i.e., the randomness in generating the results) to zero, so that the results are replicable.²⁷ To compare similarities, we first convert reviews into embeddings using the benchmark model all-MiniLM-L6-v2, and then calculate the cosine similarities of the embeddings of reviews within the same product. We also try other embedding models as a robustness check in Appendix A.1.

The results are shown in Panel A of Table 3. The p-values are from Welch’s t-tests using fake reviews as the reference group. We find that fake reviews are significantly more similar to each other, express significantly more positive sentiments, and are significantly longer with more words and more characters. All the differences are statistically significant regardless of whether we compare fake reviews with all authentic reviews or authentic five-star reviews. These results replicate the findings in the literature and support the intuition for the overrepresentation of fake reviews discussed in Section 4.2.

6.1.2 Empirical Strategy

We test Hypothesis 1 here. Overrepresentation of fake reviews, as defined in Definition 1 (i.e., $\mathbb{E}[W_{ij} \mid F_{ij} = 1] > \mathbb{E}[W_{ij} \mid F_{ij} = 0]$), involves the joint distribution of two random variables F_{ij} (the indicator of whether review i of product j is fake) and W_{ij} (the number of extracted keywords

that review i of product j mentions). However, we cannot directly observe F_{ij} or W_{ij} , because fake reviews are not easily observable and we do not know exactly which reviews contribute to each AI summary keyword.

We already have an approximation for F_{ij} in data source 2.²³ We take two steps to approximate W_{ij} , i.e., the number of extracted keywords mentioned by review i of product j . In the first step, we get the list of extracted keywords; in the second step, we determine how many extracted keywords in the list each review mentions.

Step 1 (Getting the list of extracted keywords). The most straightforward way is to directly use the list of keywords extracted by Amazon and displayed on its interface. We also use BERTopic (Grootendorst, 2022) and Top2Vec (Angelov, 2020) to extract keywords from the reviews in our data set. They both use UMAP to reduce the dimensionality of the embeddings (generated by all-MiniLM-L6-v2) and HDBSCAN to cluster the embeddings. After that, BERTopic uses TF-IDF to identify salient n-grams to assign as the keyword to each cluster of embeddings, while Top2Vec constructs a topic vector as the centroid of the review embeddings for each cluster, and then ranks candidate words and phrases by their cosine similarity to this topic vector. In each method, we retain the top representative phrase for each topic after filtering out generic marketplace terms and near-duplicate expressions, yielding a product-specific keyword set.

Step 2 (Determining how many extracted keywords each review mentions). We use two complementary criteria. First, we apply case-insensitive whole-phrase lexical matching. Second, we apply sentence-level semantic matching: each review is split into sentences, and a keyword is counted as mentioned if the cosine similarity between the keyword embedding and at least one sentence embedding in the review exceeds a fixed threshold. We use threshold 0.8 in the main text and report the analysis with other thresholds in Appendix A.2 for a robustness check. A keyword is treated as mentioned if either the lexical or semantic criterion is satisfied. Finally, we count the number of distinct extracted keywords mentioned by review i of product j , yielding three constructions of W_{ij} .

6.1.3 Results

The results of the comparisons between fake reviews and authentic reviews are reported in Panel B of Table 3. The p-values come from Welch’s t-tests using fake reviews as the reference group. We report all the results in Panel B using other embeddings as a robustness check in Appendix A.1. The scale varies, because TF-IDF in BERTopic selects words that are mentioned the most frequently, while Top2Vec selects words that match the centroid of embeddings. No matter which method we use, the average W_{ij} of fake reviews is significantly larger than the average W_{ij} of authentic reviews, indicating that $\mathbb{E}[W_{ij}|F_{ij} = 1] > \mathbb{E}[W_{ij}|F_{ij} = 0]$. This supports the overrepresentation of fake reviews.

One concern with the empirical analysis above is that some reviews may have been removed between when data source 2 was collected and when the keyword list was compiled, meaning the keywords are extracted from a slightly different set of reviews. However, the salient common themes that consumers discuss for a given product tend to be stable across review samples. For instance, 57.14% of products in Automotive have the keyword “Ease of installation”, and 89.86% of products in Grocery & Gourmet Food have the keyword “Taste”. This effect is also strong across categories. Many of the extracted keywords are common topics that reviewers of different products usually discuss. 77.23% of products in all categories have the keyword “Value for money”, and 63.27% of products in all categories have the keyword “Quality”. The common topics are common across categories and products, suggesting that they are stable and do not vary a lot across which specific reviews are sampled. Therefore, minor discrepancy in the underlying review set is unlikely to materially affect the extracted keywords. Our other analyses below can further alleviate this concern.

1. We exclude reviews that were removed by Amazon identified by comparing with the reviews documented in Hou et al. (2024). The results are reported in Panel C of Table 3. The magnitude of the difference is even larger than that reported in Panel B of Table 3.
2. We use two other methods, i.e., BERTopic and Top2Vec, to extract keywords from the reviews in our data set. The results reported in Panel B of Table 3 are also consistent.

One concern is that the way we determine whether a review mentions an extracted keyword is

different from what is actually used by Amazon. We also perform other analyses to alleviate this concern.

1. We perform the same analysis with other embedding models in Appendix A.1.
2. We perform the same analysis with other thresholds above which reviews are classified to mention one keyword in Appendix A.2.
3. We try a third method that does not involve lexical or semantic matching to construct W_{ij} . We cut reviews into sentences, convert sentences to embeddings, use UMAP to reduce the dimensionality of the embeddings, and run HDBSCAN to cluster the embeddings of sentences. Finally, we count the number of sentences that fall into clusters for each review and use it as W_{ij} . The results are reported in Panel C of Table 3.

To further alleviate remaining concerns, we perform a product-level analysis on the real output of the AI summarization algorithms deployed by Amazon on all consumer reviews. The results are reported in Section 6.2.

Table 3: Review-Level Analysis

	Fake Reviews	Authentic Reviews	Authentic 5-Star Reviews
Panel A: Textual Features of Fake Review VS Authentic Reviews			
Length (# Characters)	302.30	230.48 ($p < 0.001$)	244.64 ($p < 0.001$)
Length (# Words)	57.66	46.02 ($p < 0.001$)	43.43 ($p < 0.001$)
Sentiment	4.72		4.66 ($p < 0.001$)
Cosine Similarity	0.4287	0.2993 ($p < 0.001$)	0.3708 ($p < 0.001$)
Panel B: Testing Hypothesis 1 by Comparing # Extracted Keywords Mentioned by Fake Reviews VS Authentic Reviews			
# Keywords Extracted by Amazon	0.4017	0.2345 ($p < 0.001$)	0.3328 ($p < 0.001$)
# Keywords Extracted by BERTopic	0.7132	0.5268 ($p < 0.001$)	0.5119 ($p < 0.001$)
# Keywords Extracted by Top2Vec	0.1765	0.0993 ($p < 0.001$)	0.1238 ($p < 0.001$)
Panel C: Robustness Check			
Using HDBSCAN to Decide Mentions	2.3273	1.4859 ($p < 0.001$)	1.5821 ($p < 0.001$)
Excluding Reviews Removed by Amazon	0.4298	0.2390 ($p < 0.001$)	0.3374 ($p < 0.001$)

Note. This table compares fake reviews with all authentic reviews and with authentic 5-star reviews in the archived review dataset. Reported p-values are from Welch’s t-tests using fake reviews as the reference group. In Panel A, review length is measured by the number of characters and words in the review text; sentiment is scored by OpenAI GPT-4.1; and cosine similarity is the average within-product pairwise cosine similarity of review embeddings. Panel B tests Hypothesis 1 by comparing W_{ij} , i.e., the number of extracted keywords, mentioned by fake reviews versus authentic reviews. W_{ij} is constructed in two steps. First, for each product, we obtain a list of keywords by collecting from the interface of Amazon, or running BERTopic or Top2Vec. Second, we determine how many extracted keywords each review mentions. Panel C reports robustness checks. In the first row, mentions are determined using HDBSCAN-based clustering of sentence embeddings, and W_{ij} is defined as the number of clustered sentences in each review. In the second row, reviews later removed by Amazon are excluded from the analysis.

6.2 Product-Level Analysis

6.2.1 Empirical Strategy

We turn to product-level analysis with the observable output of real algorithms deployed by Amazon on all consumer reviews. However, on Amazon, we can only observe product-level data. With observables publicly available on the Amazon interface, we can estimate $w_j \equiv \mathbb{E}[W_{ij}|j]$, which is the mean of W_{ij} for product j , using the sample average $\hat{w}_j$, defined as the summation of the number of reviews that mention each keyword (i.e., the total number of mentions of all keywords) divided by the total number of reviews. The number of reviews that mention each keyword and the total number of reviews are downloaded directly from Amazon (data source 1 in Section 5).

We can also identify products with a higher share of fake reviews (product j with higher $f_j \equiv \mathbb{E}[F_{ij} | j]$).

1. **Amazon-sold products:** Amazon very rarely participates in review manipulation (He et al., 2022b), so we use Amazon-sold products as a proxy for products with a lower share of fake reviews (lower f_j).
2. **Suspicious products classified by RateBud:** We collect suspicious products from RateBud,⁷ a commercial website that evaluates the credibility of reviews of products on Amazon. RateBud provides letter grades for review credibility summarized in Table 2. A and B are largely free from fraud, while D, E, and F are suspected manipulators. C, in the middle, is somewhat ambiguous. In the main text, we regard products in C, D, E, and F as a subgroup of products with a higher share of fake reviews (higher f_j). We report the results when we regard products in D, E, and F as a subgroup of products with a higher share of fake reviews (higher f_j) in Appendix A.3. The results are largely consistent.

If Hypothesis 1 holds, we should expect that W_{ij} and F_{ij} are also positively correlated on the product level, i.e., products with higher $f_j \equiv \mathbb{E}[F_{ij} | j]$ also have higher $w_j \equiv \mathbb{E}[W_{ij} | j]$. We compare the average number of keywords represented in AI summaries ($\hat{w}_j$) between products with a higher share of fake reviews (higher f_j) and a lower share of fake reviews (lower f_j). We formally discuss the gap between the empirical evidence on the product level and the overrepresentation of fake reviews defined on the review level in Section 6.2.3.

6.2.2 Results

Table 4 presents the results. We compare Amazon-sold products and non-Amazon products in Column (1), and fake review products classified by RateBud and other products in Column (2). In Column (3), we conduct a stricter comparison. We compare fake review products classified by RateBud with the other products within non-Amazon products. The results consistently show that W_{ij} (the number of extracted keywords mentioned by reviews) and F_{ij} (indicator for whether a review is fake) are positively correlated on the product level.

Table 4: Product-Level Analysis

Sample	(1) All	(2) All	(3) All	(4) Unbounded	(5) Unbounded	(6) Unbounded	(7) All	(8) All	(9) All
Amazon-Sold	-0.126*** (0.00683)			-0.0611*** (0.0153)			-0.0999*** (0.00700)		-0.0974*** (0.00697)
Classified as Fake		0.351*** (0.0152)	0.319*** (0.0179)		0.242*** (0.0204)	0.216*** (0.0231)		0.215*** (0.0133)	0.211*** (0.0132)
# Keywords							0.0233*** (0.00206)	0.0264*** (0.00207)	0.0276*** (0.00206)
Avg Rating							-1.364*** (0.0578)	-1.270*** (0.0584)	-1.267*** (0.0580)
# Reviews							-1.50e-06*** (1.18e-07)	-1.70e-06*** (1.20e-07)	-1.55e-06*** (1.19e-07)
Variance of Ratings							-0.171*** (0.0125)	-0.154*** (0.0127)	-0.156*** (0.0126)
Share of 5-Star Reviews							3.563*** (0.115)	3.406*** (0.116)	3.391*** (0.115)
Constant	0.454*** (0.00387)	0.397*** (0.00326)	0.437*** (0.00423)	0.354*** (0.00744)	0.312*** (0.00662)	0.328*** (0.00785)	3.922*** (0.198)	3.538*** (0.200)	3.560*** (0.199)
Search Term $\times$ Page No. FE	No	No	No	No	No	No	Yes	Yes	Yes
Category FE	No	No	No	No	No	No	Yes	Yes	Yes
Observations	14416	14284	9658	2537	2528	1916	13477	13345	13323
R^2	0.023	0.036	0.032	0.006	0.053	0.044	0.509	0.512	0.520

Note: The DV is $\hat{w}_i$, computed as the total number of keyword mentions divided by the total number of reviews. “Amazon-Sold” proxies for products with lower share of fake reviews, while “Classified as Fake” proxies for products with higher share of fake reviews based on RateBud review credibility grades. Columns (1)–(3) use the full sample, Columns (4)–(6) restrict attention to products not bounded by the maximum number of displayed keywords (# extracted keywords smaller than 8), and Columns (7)–(9) further include search term $\times$ page number and category fixed effects.

Standard errors are in parentheses; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

One concern with this analysis is that there may be an upper bound on the number of keywords that can be displayed on the Amazon interface. Products with a lower share of fake reviews may be more popular with a larger number of reviews, but their number of keywords is bounded. Therefore, some of their keywords are dropped, which can explain why on average their reviews mention fewer keywords. In Columns (4)–(6) of Table 4, we replicate the analysis on the subset of products that are not bounded by the upper bound (the number of keywords is smaller than the

maximum number of keywords in the dataset). The results are largely consistent, suggesting that this explanation is at least not the whole story.

One may also question whether the product-level correlation is due to other differences of products with a higher versus lower share of fake reviews. For example, they may belong to totally different categories. To mitigate this concern, we control for categories and search term $\times$ page number fixed effects. They are included to control for unobservable characteristics of the products and to ensure that our comparison is within the products shown on the same search result page. We also control other observables. The results are consistent. However, this may not fully rule out the possibilities that overrepresentation of fake reviews does not hold ($\mathbb{E}[W_{ij} | F_{ij} = 1] \leq \mathbb{E}[W_{ij} | F_{ij} = 0]$), but other factors lead to the product-level positive correlation of W_{ij} and F_{ij} . We formally discuss this issue in Section 6.2.3.

Other concerns include that the results in Table 4 are driven by the fact that we normalize the metrics by dividing the total number of reviews, and that Amazon may only select a fixed number of reviews to extract keywords from. Please see Appendix A.4 for a detailed discussion.

6.2.3 Discussion

We study in what sense the product-level evidence can support the overrepresentation of fake reviews defined at the review level. For the ease of exposition, we introduce some new notations. We consider two subgroups of products J_1, J_2 that partition the set of products. Products in J_1 have a higher share of fake reviews than those in J_2 , i.e. $\mathbb{E}[f_j | j \in J_1] > \mathbb{E}[f_j | j \in J_2]$. We have empirically found that $\mathbb{E}[w_j | j \in J_1] > \mathbb{E}[w_j | j \in J_2]$. We define $w_{F,j} \equiv \mathbb{E}[W_{ij} | F_{ij} = 1, j]$, $w_{A,j} \equiv \mathbb{E}[W_{ij} | F_{ij} = 0, j]$, and $\delta_j \equiv w_{F,j} - w_{A,j}$.

We first decompose $\mathbb{E}[w_j | j \in J_1] - \mathbb{E}[w_j | j \in J_2]$ in Equation (1), where $\bar{\delta} \equiv \mathbb{E}[\delta_j]$ is the mean of δ_j across products. We would like to infer whether $\bar{\delta} > 0$. In other words, we aim to study whether on average $\delta_j \equiv \mathbb{E}[W_{ij} | F_{ij} = 1, j] - \mathbb{E}[W_{ij} | F_{ij} = 0, j]$ is greater than 0. There are two heterogeneity terms (B and H) that can also influence $\mathbb{E}[w_j | j \in J_1] - \mathbb{E}[w_j | j \in J_2]$. B is due to the

heterogeneity in the baseline $w_{A,j}$ while H is due to the heterogeneity in δ_j .

$$\begin{aligned}
\mathbb{E}[w_j \mid j \in J_1] - \mathbb{E}[w_j \mid j \in J_2] &= \mathbb{E}[w_{A,j}(1 - f_j) + w_{F,j}f_j \mid J_1] - \mathbb{E}[w_{A,j}(1 - f_j) + w_{F,j}f_j \mid J_2] \\
&= \bar{\delta}(\mathbb{E}[f_j \mid j \in J_1] - \mathbb{E}[f_j \mid j \in J_2]) \\
&\quad + \overbrace{\mathbb{E}[w_{A,j} \mid j \in J_1] - \mathbb{E}[w_{A,j} \mid j \in J_2]}^B \\
&\quad + \overbrace{\mathbb{E}[f_j(\delta_j - \bar{\delta}) \mid j \in J_1] - \mathbb{E}[f_j(\delta_j - \bar{\delta}) \mid j \in J_2]}^H
\end{aligned} \tag{1}$$

As a baseline reference for the analysis below, according to Proposition 1, if we assume no heterogeneity in the mean of W_{ij} conditional on F_{ij} (i.e., $\mathbb{E}[W_{ij} \mid F_{ij}, j] = \mathbb{E}[W_{ij} \mid F_{ij}]$), our empirical results are equivalent to proving the overrepresentation of fake reviews defined at the review level. The intuition is straightforward: once heterogeneity in the mean is shut down, evidence at the aggregate level contains most of the individual-level information.

Proposition 1. Assuming $\mathbb{E}[W_{ij} \mid F_{ij}, j] = \mathbb{E}[W_{ij} \mid F_{ij}]$, $\mathbb{E}[W_{ij} \mid F_{ij} = 1] > \mathbb{E}[W_{ij} \mid F_{ij} = 0]$ if and only if $\mathbb{E}[w_j \mid j \in J_1] > \mathbb{E}[w_j \mid j \in J_2]$.

Proof. Without heterogeneity in the mean, $B = H = 0$. By definition, $\mathbb{E}[f_j \mid j \in J_1] - \mathbb{E}[f_j \mid j \in J_2] > 0$, so our empirical finding that $\mathbb{E}[w_j \mid j \in J_1] - \mathbb{E}[w_j \mid j \in J_2] > 0$ is equivalent to $\bar{\delta} > 0$, which directly means that $\mathbb{E}[W_{ij} \mid F_{ij} = 1] - \mathbb{E}[W_{ij} \mid F_{ij} = 0] > 0$. $\square$

However, B and H are typically nonzero. Hereafter, we attempt to study how much of $\mathbb{E}[w_j \mid j \in J_1] - \mathbb{E}[w_j \mid j \in J_2]$ can be explained by heterogeneity in the mean of W_{ij} across products rather than the overrepresentation of fake reviews. In other words, we study the scale of B and H .

Estimating B . We take advantage of the nature of the market for fake reviews. Amazon has implemented relatively strict regulation on review manipulation. Therefore, sellers who want to buy fake reviews typically have to cover the costs of fake reviewers to buy the product to ensure that they are verified purchasers.¹⁶ This implies that the cost of buying fake reviews increases with the price of the products. Therefore, by directly observing review manipulation activities, He et al. (2022b) notice that there are far fewer expensive products among fake review products than among their competitors. This finding is consistent with the report by the Wall Street Journal

that raises cheap “five-star” no-name products as the example for typical review manipulators,²⁸ and the report by Wired that uses “low-cost tech products” as the typical example of review manipulators.²⁹ Additionally, according to Saoud Khalifah, the founder of Fakespot, the main buyers of fake reviews are sellers of low-price products (around \$15 - \$40).¹⁶ These imply that $w_j \approx w_{A,j} \equiv \mathbb{E}[W_{ij} \mid F_{ij} = 0, j]$ among expensive products.

In Figure 3, we plot the difference (Non-Amazon – Amazon-Sold Products) in $\hat{w}_j$ on the subsample of products priced higher than the threshold on the horizontal axis. The difference in $\hat{w}_j$ between non-Amazon and Amazon-sold products vanishes with a higher price threshold. In other words, without the difference in the percentage of fake reviews between Amazon-sold and non-Amazon products (since expensive products typically do not have fake reviews), there is almost no difference in $\hat{w}_j$, suggesting that the difference in the percentage of fake reviews, rather than other differences between Amazon-sold and non-Amazon products, drives the empirical finding.

More importantly, Figure 3 can help decompose the observed difference $\mathbb{E}[w_j \mid j \in J_1] - \mathbb{E}[w_j \mid j \in J_2]$. The difference decreases until it reaches a plateau at the price threshold of about \$230. We use $\mathbb{E}[w_j \mid j \in J_1] - \mathbb{E}[w_j \mid j \in J_2]$ in the subsample of products priced higher than \$250 to approximate $B \equiv \mathbb{E}[w_{A,j} \mid j \in J_1] - \mathbb{E}[w_{A,j} \mid j \in J_2]$. The result is shown in Table 5. In Appendix B.2, we also consider a worse case in which we use the upper endpoint of the 95% confidence interval to approximate B .

H cannot be estimated from the data, so we have to derive a loose worst-case bound of $|H|$.

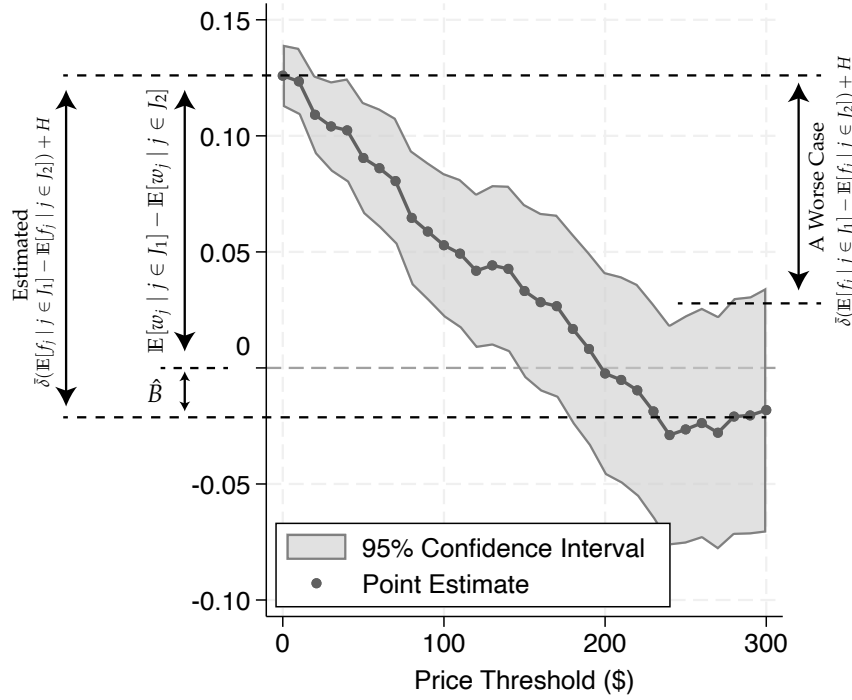
Lemma 1. *For any random variable Z_j , the following inequality holds, with equality achieved when $\mathbb{V}(Z_j \mid j \in J_1) = \mathbb{V}(Z_j \mid j \in J_2) = 0$.*

$$|\mathbb{E}[Z_j \mid j \in J_1] - \mathbb{E}[Z_j \mid j \in J_2]| \leq \sqrt{\frac{\mathbb{V}(Z_j)}{p(j \in J_1)p(j \in J_2)}}$$

Proof. See Appendix B.1. □

We define f_{WRMS} as the weighted root mean square of f_j , with weights $(\delta_j - \bar{\delta})^2$, i.e., $f_{WRMS}^2 \equiv \frac{\mathbb{E}[f_j^2(\delta_j - \bar{\delta})^2]}{\mathbb{E}[(\delta_j - \bar{\delta})^2]}$. We also define $\bar{f}$ as the weighted average of f_j , with weights $(\delta_j - \bar{\delta})^2$, i.e., $\bar{f} \equiv \frac{\mathbb{E}[f_j(\delta_j - \bar{\delta})^2]}{\mathbb{E}[(\delta_j - \bar{\delta})^2]}$.

Figure 3: Decomposition of $\mathbb{E}[w_j | j \in J_1] - \mathbb{E}[w_j | j \in J_2]$ with J_1 Denoting Non-Amazon Products And J_2 Denoting Amazon-Sold Products



It is easy to show that $f_{WRMS}^2 \leq \bar{f}$. According to Lemma 1,

$$|H| \leq \sqrt{\frac{\mathbb{V}(f_j(\delta_j - \bar{\delta}))}{p(j \in J_1)p(j \in J_2)}} = \sqrt{\frac{\mathbb{E}[f_j^2(\delta_j - \bar{\delta})^2] - \mathbb{C}(f_j, \delta_j)^2}{p(j \in J_1)p(j \in J_2)}} \leq f_{WRMS} \sqrt{\frac{\mathbb{V}(\delta_j)}{p(j \in J_1)p(j \in J_2)}}$$

We can easily estimate $p(j \in J_1)$ and $p(j \in J_2)$. We can also estimate $\mathbb{V}(\delta_j)$ in review-level data from data source 2. The details are elaborated in Appendix B.3. Then, we can estimate $\sqrt{\mathbb{V}(\delta_j)/p(j \in J_1)/p(j \in J_2)}$. The result is shown in Table 5.

We make a null hypothesis that $\bar{\delta} \leq 0$, i.e., the overrepresentation of fake reviews is not the explanation of our empirical finding that $\mathbb{E}[w_j | j \in J_1] > \mathbb{E}[w_j | j \in J_2]$. If the null hypothesis is true, in the worst case with $\mathbb{V}(f_j(\delta_j - \bar{\delta}) | j \in J_1) = \mathbb{V}(f_j(\delta_j - \bar{\delta}) | j \in J_2) = 0$ and $\mathbb{C}(f_j, \delta_j) = 0$, we need f_{WRMS} to be at least 56.00% ($\bar{f}$ to be at least 31.36%). Given that around 30% of products are directly sold by Amazon, it is implausible that on average, among products in top search results, more than one third of the reviews are fake. Therefore, the null hypothesis is not plausible, so $\bar{\delta} > 0$ is more plausible.

Table 5: Decomposition of $\mathbb{E}[w_j | j \in J_1] - \mathbb{E}[w_j | j \in J_2]$ with J_1 Denoting Non-Amazon Products And J_2 Denoting Amazon-Sold Products

	Quantity	Estimate
	Estimated $\mathbb{E}[w_j j \in J_1] - \mathbb{E}[w_j j \in J_2]$	0.1259
–	Estimated $B \equiv \mathbb{E}[w_{A,j} j \in J_1] - \mathbb{E}[w_{A,j} j \in J_2]$	-0.0265
=	Estimated $\hat{\delta}(\mathbb{E}[f_j j \in J_1] - \mathbb{E}[f_j j \in J_2]) + H$	0.1524
÷	Estimated $\sqrt{\mathbb{V}(\delta_j)/p(j \in J_1)/p(j \in J_2)}$	0.2722
=	Smallest f_{WRMS} to rule out overrepresentation of fake reviews in the worst case	56.00%

7 Effects of Overrepresentation

7.1 Effect 1: Distortion in Product Quality Information

We test Hypothesis 2, which states that the sentiment of AI summaries is biased in favor of review manipulators. Specifically, we expect that products with a higher share of fake reviews have: (i) more positive sentiment labels corresponding to extracted keywords, and (ii) more positive AI summary paragraphs.

7.1.1 Empirical Strategy

The general idea of the empirical analysis is straightforward. We want to compare the two dependent variables between products with a higher (higher $f_j \equiv \mathbb{E}[F_{ij} | j]$) and lower share of fake reviews (lower f_j): sentiment labels associated with keywords and sentiments of AI summary paragraphs.

We continue to use the two proxies introduced in Section 6.2.1 to identify products with a higher share of fake reviews, i.e., non-Amazon products and products with low grades from RateBud. The sentiment label associated with each keyword is directly observable on Amazon (the checkmark and minus signs before each keyword shown in Figure 1). We assign 1 to positive sentiment (checkmark sign), -1 to negative sentiment (forbidden sign), and 0 to neutral sentiment (no sign or a short gray line). Then, we take the average across different keywords. To quantify the sentiments of AI summary paragraphs, we use two language models: OpenAI GPT-4.1²⁶ and bert-base-multilingual-uncased-sentiment.³⁰ When using OpenAI GPT-4.1, we prompt the model to predict the quality perceived by consumers who see the summary paragraph, on the scale of average ratings.³¹ bert-base-multilingual-uncased-sentiment is a specialized language model finetuned on the task of predicting the ratings of reviews. We report the results with OpenAI

GPT-4.1 in the main text, and the results with bert-base-multilingual-uncased-sentiment in Appendix C.1.

The specification we estimate is shown in Equation (2).

$$DV_j = \beta I_j + \text{Rating_Controls}_j \gamma + \phi_c + \psi_{kn} + \epsilon_j \quad (2)$$

where j denotes products, c denotes categories, k denotes search terms, and n denotes the page number in the search results on which product j is shown. DV_j is one of the two dependent variables that we just mentioned, including sentiment labels corresponding to keywords and sentiments of AI summary paragraphs. I_j is the indicator for subgroups of products with higher (or lower) share of fake reviews based on the two proxies discussed in Section 6.2.1. Rating_Controls_j is a vector of flexible controls of the rating distribution. ϕ_c is category fixed effects and ψ_{kn} is search term $\times$ page number fixed effects. They are included to control for unobservable characteristics of the products and to ensure that our comparison is within the products shown on the same search result page.

7.1.2 Results

The results are shown in Table 6. We consider I_j to indicate whether product j is sold by Amazon in Columns (1)-(4) and to indicate whether product j is a fake review product classified by RateBud in Columns (5)-(8). We group C, D, E, and F together as classified fake review products in Table 6. In Appendix C.3, we report the results with only D, E and F as classified fake review products as a robustness check.

The results of the sentiments corresponding to keywords are reported in Columns (1), (2), (5) and (6), while the results of the sentiments of AI summary paragraphs are reported in Columns (3), (4), (7) and (8). We find that with search term $\times$ page number fixed effects and category fixed effects controlled, Amazon-sold products (with lower share of fake reviews) have significantly more negative sentiment labels associated with keywords, and significantly more negative sentiments of summary paragraphs than the other products on the same search result page. The results on the classified fake review products are exactly the opposite. R^2 of the regressions is considerable, suggesting that we have controlled for many factors that can influence the dependent variables. The results validate Hypothesis 2.

Table 6: Distortion in Product Quality Information

	(1) Keyword	(2) Keyword	(3) Paragraph	(4) Paragraph	(5) Keyword	(6) Keyword	(7) Paragraph	(8) Paragraph
Amazon-Sold	-0.0469*** (0.00475)	-0.0450*** (0.00476)	-0.0861*** (0.00882)	-0.0807*** (0.00883)				
Classified as Fake					0.0440*** (0.00900)	0.0473*** (0.00902)	0.116*** (0.0167)	0.112*** (0.0167)
Avg Rating	0.692*** (0.00846)	0.835*** (0.0393)	1.165*** (0.0157)	1.047*** (0.0729)	0.694*** (0.00854)	0.864*** (0.0399)	1.172*** (0.0158)	1.121*** (0.0740)
# Reviews		-2.36e-07** (7.96e-08)		-6.64e-07*** (1.48e-07)		-3.25e-07*** (8.14e-08)		-8.40e-07*** (1.51e-07)
Variance of Ratings		0.0642*** (0.00849)		0.0526*** (0.0158)		0.0719*** (0.00864)		0.0697*** (0.0160)
Share of 5-Star Reviews		-0.0271 (0.0780)		0.705*** (0.145)		-0.0583 (0.0791)		0.615*** (0.146)
Constant	-2.366*** (0.0376)	-3.067*** (0.135)	-0.973*** (0.0697)	-1.045*** (0.250)	-2.392*** (0.0379)	-3.202*** (0.137)	-1.036*** (0.0702)	-1.358*** (0.254)
Search Term $\times$ Page No. FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Category FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	13515	13515	13515	13515	13382	13382	13382	13382
R ²	0.553	0.557	0.528	0.532	0.552	0.556	0.528	0.533

Note. This table tests Hypothesis 2 by comparing sentiments of AI summaries between products with higher versus lower share of fake reviews. The dependent variables are (i) Keyword Sentiment, constructed by coding each keyword's sentiment label as 1 (positive/checkmark), 0 (neutral/no sign), or -1 (negative/minus) and averaging across keywords, and (ii) Paragraph Sentiment, measured by prompting OpenAI GPT-4.1 to predict the perceived product quality from the AI summary paragraph. As for the proxy for products more and less likely to have fake reviews, Columns (1)-(4) use an Amazon-Sold indicator; Columns (5)-(8) use a dummy variable "Classified as Fake" to indicate whether the product is of grades C/D/E/F on RateBud.

Standard errors are in parentheses; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

7.1.3 Robustness Check

Table 7 presents two robustness checks. First, we put the two group indicators, classified fake review products and Amazon-sold products, together in the same regression. Columns (1) - (4) report the results. Second, we continue the stricter comparison in Panel C of Table 4, in which we compare fake review products classified by RateBud and the other products within non-Amazon products. Columns (5) - (8) report the results. Both robustness checks yield results consistent with Table 6.

7.1.4 Discussion on the Causal Relationship

Hypothesis 2 is not a causal claim, but we would like to shed light on whether the difference in the sentiment of AI summaries is caused by the difference in the share of fake reviews between subgroups with a higher and lower share of fake reviews rather than other differences between subgroups. We use three approaches to provide suggestive evidence of a causal relationship. First,

Table 7: Robustness Check for Distortion in Product Quality Information

	(1) Keyword	(2) Keyword	(3) Paragraph	(4) Paragraph	(5) Keyword	(6) Keyword	(7) Paragraph	(8) Paragraph
Amazon-Sold	-0.0463*** (0.00478)	-0.0443*** (0.00478)	-0.0855*** (0.00886)	-0.0800*** (0.00886)				
Classified as Fake	0.0420*** (0.00897)	0.0456*** (0.00898)	0.112*** (0.0166)	0.109*** (0.0166)	0.0343*** (0.0103)	0.0382*** (0.0103)	0.0978*** (0.0190)	0.0958*** (0.0191)
Avg Rating	0.696*** (0.00852)	0.871*** (0.0398)	1.175*** (0.0158)	1.132*** (0.0738)	0.659*** (0.0105)	0.897*** (0.0503)	1.120*** (0.0195)	1.153*** (0.0931)
# Reviews		-2.59e-07** (8.14e-08)		-7.18e-07*** (1.51e-07)		-5.20e-07*** (1.27e-07)		-1.03e-06*** (2.36e-07)
Variance of Ratings		0.0722*** (0.00861)		0.0703*** (0.0160)		0.0812*** (0.0109)		0.0850*** (0.0202)
Share of 5-Star Reviews		-0.0727 (0.0788)		0.589*** (0.146)		-0.200* (0.0966)		0.452* (0.179)
Constant	-2.386*** (0.0378)	-3.208*** (0.137)	-1.020*** (0.0701)	-1.365*** (0.253)	-2.218*** (0.0466)	-3.240*** (0.174)	-0.779*** (0.0864)	-1.376*** (0.323)
Search Term $\times$ Page No. FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Category FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	13360	13360	13360	13360	9126	9126	9126	9126
R ²	0.556	0.560	0.532	0.537	0.565	0.570	0.542	0.546

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

we flexibly control the distribution of ratings in the specification in Equation (2). Second, we control the characteristics of products by fixed effects in Equation (2), including the categories of the products and search term $\times$ page number fixed effects. In this way, we only compare among products on the same search result page. Third, we also conduct a falsification test here.

The cost of buying fake reviews increases with the price of the products because it is common for sellers to reimburse the price of the product to fake reviewers. Therefore, we expect that more expensive products are less likely to have fake reviews, and that there is no significant difference in fake reviews between expensive Amazon-sold products and expensive non-Amazon products. If the difference in fake reviews plays an important role in the results, the results should no longer hold without the difference in fake reviews among expensive products.

In Figure 8, we rerun the comparison between Amazon-sold and non-Amazon products in Table 6 in the subsample of products priced below 25% percentile or above 75% percentile. As is evident, the coefficients are significantly negative only when the prices of the products are fairly low, suggesting that overrepresentation of fake reviews is likely to be the reason underlying the difference in the sentiments between Amazon-sold products and non-Amazon products.

Table 8: Falsification Test for Distortion in Production Quality Information

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Sample	Keyword Above 75%	Keyword Above 75%	Paragraph Above 75%	Paragraph Above 75%	Keyword Below 25%	Keyword Below 25%	Paragraph Below 25%	Paragraph Below 25%
Amazon-Sold	-0.00611 (0.0109)	-0.000518 (0.0109)	-0.0317 (0.0200)	-0.0200 (0.0199)	-0.0800*** (0.00980)	-0.0786*** (0.00979)	-0.130*** (0.0184)	-0.124*** (0.0184)
Avg Rating	0.599*** (0.0167)	0.754*** (0.0733)	0.999*** (0.0305)	0.966*** (0.134)	0.875*** (0.0221)	1.053*** (0.101)	1.508*** (0.0415)	1.254*** (0.189)
# Reviews		-2.23e-06*** (4.90e-07)		-4.66e-06*** (8.99e-07)		-7.59e-08 (1.06e-07)		-2.97e-07 (1.98e-07)
Variance of Ratings		0.0804*** (0.0176)		0.0659* (0.0322)		0.0824*** (0.0181)		0.0739* (0.0340)
Share of 5-Star Reviews		0.00888 (0.148)		0.531 (0.271)		0.0373 (0.202)		1.280*** (0.380)
Constant	-1.941*** (0.0722)	-2.751*** (0.254)	-0.211 (0.132)	-0.552 (0.465)	-3.217*** (0.0996)	-4.145*** (0.337)	-2.568*** (0.187)	-2.484*** (0.633)
Search Term $\times$ Page No. FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Category FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2854	2854	2854	2854	3335	3335	3335	3335
R ²	0.606	0.614	0.567	0.575	0.580	0.586	0.570	0.576

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

7.2 Effect 2: Distortion in Market Outcomes

We test Hypothesis 3 to investigate whether, after the introduction of AI summarization, review manipulators experience greater sales than other sellers.

7.2.1 Empirical Strategy

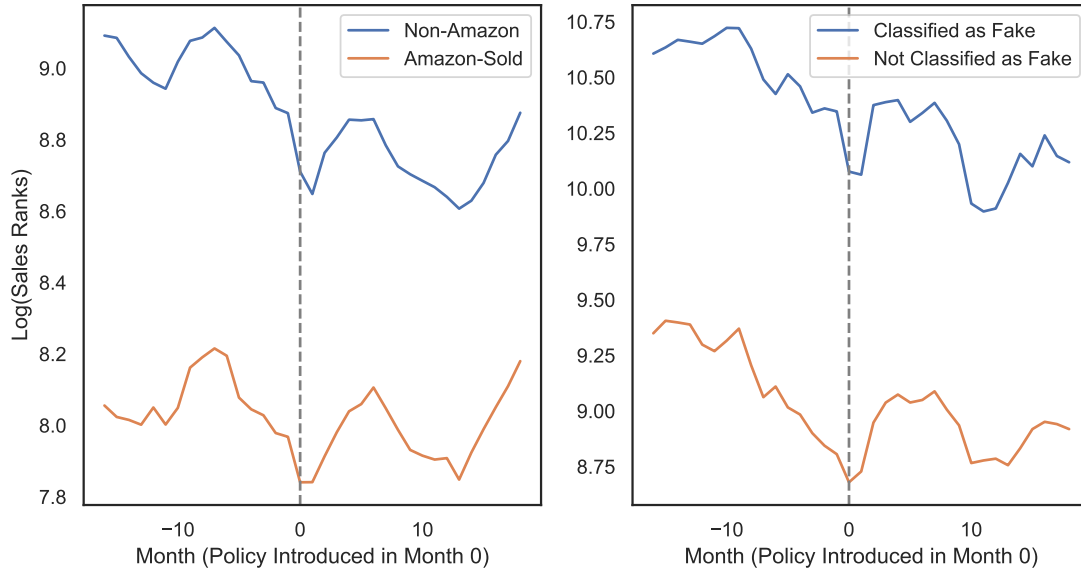
Following the classic paper in the literature on online reviews, [Chevalier and Mayzlin \(2006\)](#), we use the logarithms of sales ranks as our dependent variable. We collect sales rank history from Keepa. We select products that have observations both before and after the introduction of AI summarization. We aggregate the observations of ranks to the daily level to make the panel more balanced.

To make the subgroups of products comparable, we run exact matching on categories and the interaction of search terms and page numbers. Specifically, we first define the matching strata as category $\times$ (keyword $\times$ page). We only compare products that are in the same product category³² and appear on the same search result page (the same search term and the same search result page number). We then implement exact matching within these strata and drop observations in strata that contain products from only one group. The matching procedure produces a set of weights: products in one group receive weight one, while products in the other group are reweighted so

that, within each stratum, the sum of weights is identical across the two groups. Finally, we weight our observations with the set of weights.

We first describe the sales rank data in Figure 4. We use June 12, 2023-the date when CNBC reported that “Amazon is using generative AI to summarize product reviews”¹¹-as the introduction date. We find a parallel trend shared by different subgroups. After the introduction of AI summarization, the two blue lines (products with a higher share of fake reviews) seem to decrease more in the logarithms of sales ranks, suggesting that sales of review manipulators increase relative to their competitors after the introduction of AI summarization.

Figure 4: Description of Sales Rank Data



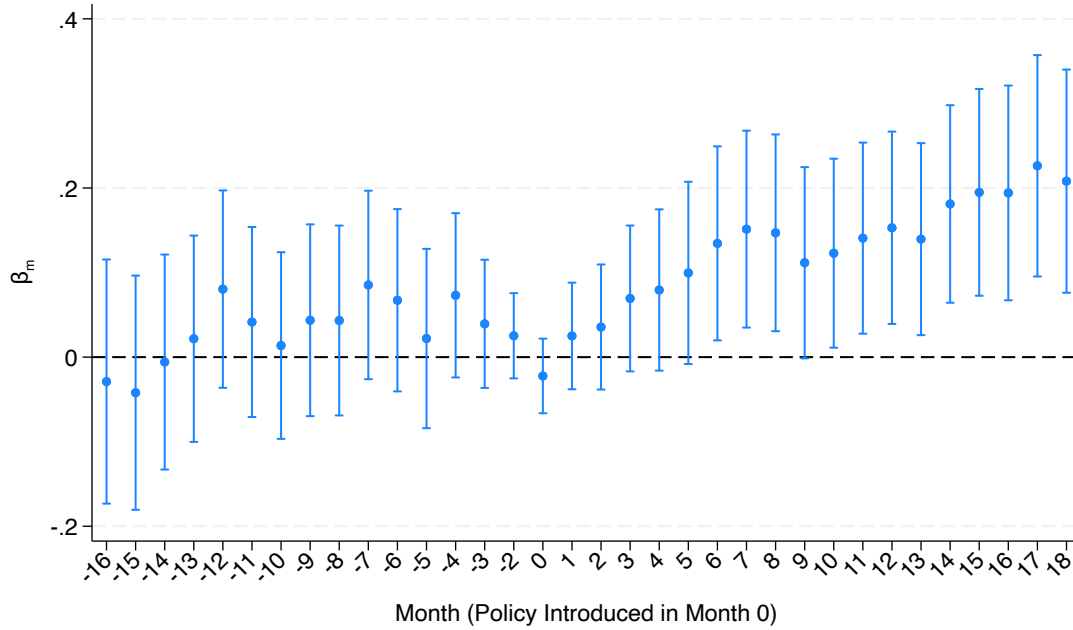
To formalize this analysis, we perform event studies in Equation (3), where j indexes products, t indexes date, and m indexes month. I_j is again the indicator for subgroups of products with a higher (higher $f_j \equiv \mathbb{E}[F_{ij} | j]$) or lower share of fake reviews (lower f_j). We continue to use the two proxies introduced in Section 6.2.1 to identify products with a higher share of fake reviews, i.e., non-Amazon products and products with low grades from RateBud. The variable of interests is β_m , which measures the difference of the two lines in Figure 4. The formal model also allows us to conduct statistical inference. We cluster the error term at the product level to correct for serial autocorrelation over time.

$$\log(\text{Sales_Ranks}_{jt}) = \sum_m \beta_m I_j \times \phi_m + \alpha_j + \omega_t + \epsilon_{jt} \quad (3)$$

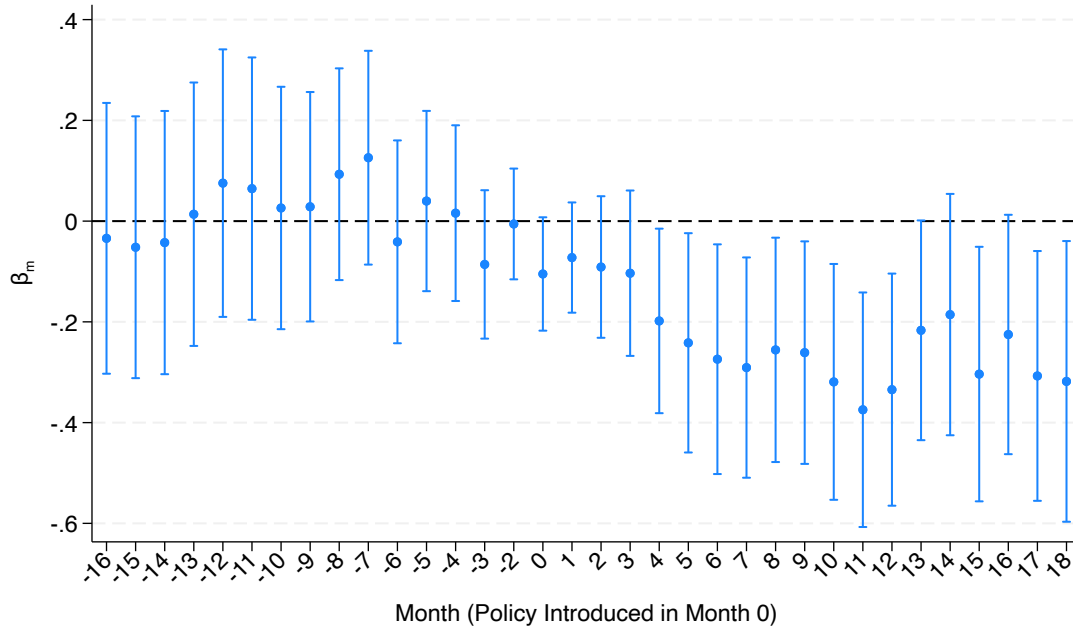
7.2.2 Results

We report the results of event studies. We plot the coefficients of the interactions of each month dummy and the group indicator. Following common practice, we normalize the month when AI summarization was introduced as Month 0 and normalize the coefficients of Month -1 to be 0. The result of the comparison between Amazon-sold and non-Amazon products is shown in Figure 5, while the result of the comparison between classified fake review products and other products is shown in Figure 6. There are no significant pretrends before the introduction of AI summarization. Instead, there is a significant break in the evolution of β_m after the introduction of AI summarization. The sales ranks of products with a higher share of fake reviews relatively decreased (i.e., moved up), suggesting that their sales relatively increased following the introduction of AI summarization.

Figure 5: Event Study for the Comparison between Amazon-Sold and Non-Amazon Products



Note: The y -axis is the estimated coefficients β_m of the regression $\log(\text{Sales_Ranks}_{jt}) = \sum_m \beta_m I_j \times \phi_m + \alpha_j + \omega_d + \epsilon_{jt}$, where ϕ_m is month dummy, I_j is the group indicator of whether the product is sold by Amazon (and therefore with a lower share of fake reviews), α_j is product dummy, and ω_d is date dummy. We plot point estimate and 95% confidence intervals. The error terms are clustered at the product level. The x -axis is m (month). This figure shows that after the introduction of AI summarization, products that are largely manipulation-free experience a relative decrease in sales. Following the introduction of AI summarization, the sales of products changed in a direction that supports the distortion in market outcomes stated in Hypothesis 3.

Figure 6: Event Study for the Comparison between Classified Fake Review Products and Other Products

Note: The y-axis is the estimated coefficients β_m of the regression $\log(\text{Sales_Ranks}_{jt}) = \sum_m \beta_m I_j \times \phi_m + \alpha_j + \omega_d + \epsilon_{jt}$, where ϕ_m is month dummy, I_j is the group indicator of whether the product is graded by RateBud as C/D/E/F, α_j is product dummy, and ω_d is date dummy. We plot point estimate and 95% confidence intervals. The error terms are clustered at the product level. The x-axis is m (month). This figure shows that after the introduction of AI summarization, products with a higher share of fake reviews experience a relative increase in sales.

8 Conclusion

Fake reviews have been shown to harm consumer welfare (He et al., 2022b; Gandhi et al., 2024), and have received much attention from the Federal Trade Commission (FTC) of the US⁵ and the Parliament of the UK.⁶ In this paper, we find evidence that the negative effects of fake reviews can be amplified by AI summarization of reviews, which has been a common practice on digital platforms. This AI bias is fundamentally rooted in the design of AI summarization algorithms. AI summarization algorithms are designed to summarize common themes among consumer-generated reviews, while fake reviews disproportionately concentrate on common themes.

We first define the overrepresentation of fake reviews. In the empirical exercise, we confirm that AI summaries overrepresent fake reviews through review-level analysis and product-level analysis. Next, we empirically test two effects of overrepresentation. First, we show that the sentiments associated with keywords and AI summary paragraphs are more positive for products with a higher share of fake reviews, even after we control for flexible metrics of reviews and abundant fixed effects. Second, we find that the introduction of AI summarization benefited review manipulators

more than the other sellers.

Although our empirical analyses focus on the popular e-commerce platform, Amazon, we believe our findings are generalizable to other digital platforms that also deploy AI summarization that extracts common themes from consumer reviews. As highlighted in Section 3, Apple App Store also selectively extracts opinions only from reviews that mention popular topics.¹⁵ Because fake reviews rely heavily on repetitive templates or coordinated campaigns instead of idiosyncratic personal experiences, they naturally form clusters around common themes. This pattern should exist across contexts. As long as AI summarization prioritizes extracting salient and frequent themes, it is likely to remain biased toward fake reviews. We leave the investigation of the overrepresentation of fake reviews on other platforms to future research.

This distortive effect on the market is particularly concerning to society. Researchers have shown that AI summaries can increase purchase rates (Wang et al., 2025) and hence benefit platforms. This can explain why platforms are increasingly adopting AI summarization. However, our findings suggest that AI summaries might direct consumers to suboptimal products due to the interaction between the design of the algorithm and the nature of fake reviews. It is plausible that this might hurt consumers. Investigating its impact on consumer welfare is an important research question that we leave for future research.

In addition, the findings are worrying for platforms that care about the long-term trust of customers and also have interests in deploying AI tools. We find that current AI summarization overrepresents fake reviews. In the long run, this might hurt customer trust in the review system. Therefore, a responsible platform that truly cares about long-term customer satisfaction should be alert to the unexpected AI bias of its algorithms. When deploying trendy AI tools on their platforms, managers should be aware that these AI tools are still evolving, and their behavior can be difficult to anticipate. They may lead to unexpected and unintended outcomes that drive customers away.

It is challenging to design AI summarization algorithms that can summarize honest common themes while avoiding common themes in false information. Simply improving the accuracy of the current algorithms only makes the bias even worse. Nevertheless, we hope this research will inspire more researchers in different fields to work towards algorithms proficient in summarizing information contaminated by misinformation, such as online reviews.

Notes

- ¹ <https://www.aboutamazon.com/news/amazon-ai/amazon-improves-customer-reviews-with-generative-ai>
- ² <https://www.wsj.com/tech/personal-tech/fake-reviews-and-inflated-ratings-are-still-a-problem-for-amazon-11623587313>
- ³ <https://apnews.com/article/fake-online-reviews-generative-ai-40f5000346b1894a778434ba295a0496>
- ⁴ <https://www.wsj.com/us-news/youre-probably-falling-for-fake-product-reviews-b4d07f23>
- ⁵ <https://www.ftc.gov/news-events/news/press-releases/2024/08/federal-trade-commission-announces-final-rule-banning-fake-reviews-testimonials>
- ⁶ <https://www.legislation.gov.uk/ukpga/2024/13/contents>
- ⁷ <https://www.ratebud.ai/>
- ⁸ Screenshot captured from <https://www.amazon.com/dp/B0DWJCK9LZ/> on November 1, 2025
- ⁹ Screenshot captured from <https://www.bestbuy.com/product/apple-11-inch-ipad-air-m3-chip-built-for-apple-intelligence-wi-fi-128gb-blue/JJGCQ8VZVZ> on November 1, 2025
- ¹⁰ <https://www.emarketer.com/content/5-charts-search-2024-google-ai-retail-media>
- ¹¹ <https://www.cnbc.com/2023/06/12/amazon-is-using-generative-ai-to-summarize-product-reviews.html>
- ¹² <https://www.cbsnews.com/news/amazon-artificial-intelligence-summarize-product-reviews/>
- ¹³ <https://aws.amazon.com/blogs/machine-learning/going-beyond-ai-assistants-examples-from-amazon-com-reinventing-industries-with-generative-ai/>
- ¹⁴ <https://www.channelmax.net/article/amazon-introduces-ai-generated-review-summaries-to-help-customers-shop-faster-and-smarter>
- ¹⁵ <https://machinelearning.apple.com/research/app-store-review>
- ¹⁶ <https://nymag.com/intelligencer/2022/07/amazon-fake-reviews-can-they-be-stopped.html>
- ¹⁷ We download the price of the default size and color. We use the post-discount price throughout the paper.
- ¹⁸ Amazon interface displays a multi-level category. When we control categories using fixed effects, we control the finest level of categories, so that we control enough unobservable confounders. When we conduct exact matching on categories, we match on the first-level category, so that we do not have to drop too many observations.
- ¹⁹ <https://explodingtopics.com/blog/most-searched-items-on-amazon>
- ²⁰ <https://meetglimpse.com/top-searched/most-searched-products-on-amazon/>
- ²¹ We also stop collecting data when we have already collected more than 110 products but not reached page seven.
- ²² Screenshot captured from <https://www.amazon.com/dp/B0B74R1VKS/> on March 3, 2026
- ²³ <https://github.com/bretthollenbeck/fake-reviews-data>
- ²⁴ <https://keepa.com/>
- ²⁵ <https://www.aboutamazon.com/news/retail/amazon-ai-generated-review-highlights>
- ²⁶ <https://openai.com/index/gpt-4-1/>
- ²⁷ The prompt we use is as follows. System prompt: “You are an AI assistant that scores the sentiment of reviews on Amazon. Respond succinctly in the requested format.” User prompt: “Analyze the sentiment of reviews. What is the quality perceived by consumers? Please output a float number: X. X should be between 1 and 5. Review Text: {Review Text}”
- ²⁸ <https://www.wsj.com/tech/personal-tech/how-to-spot-fake-reviews-and-shady-ratings-on-amazon-11658333513>
- ²⁹ <https://www.wired.com/story/bug-zappers-are-swarming-on-amazon/>
- ³⁰ <https://huggingface.co/nlptown/bert-base-multilingual-uncased-sentiment>
- ³¹ The prompt we use is as follows. System prompt: “You are an AI assistant that scores the sentiment of AI summary of online reviews on Amazon. Respond succinctly in the requested format.” User prompt: “Analyze the sentiment of AI summary of reviews. What is the quality perceived by consumers? Please output a float number: X. X should be between 1 and 5. AI Summary: {The content of AI summary}”
- ³² Here, we only consider the first-level category to avoid dropping too many observations.

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Appendix A Robustness Checks for Overrepresentation of Fake Reviews

A.1 Other Embedding Models

We rerun the analyses with a larger embedding model named `paraphrase-mpnet-base-v2`. The results are shown in Table A.1. The results are consistent with those reported in Table 3 in the main text.

A.2 Other Thresholds

When determining how many extracted keywords each review mentions, we use a semantic matching that treats a keyword as mentioned if the cosine similarity between the keyword embedding and at least one sentence embedding in the review exceeds a threshold. In the main text, we set the threshold to be 0.8. Here, we report the results with the threshold set to be 0.6 and 0.4. The results with the thresholds 0.6 and 0.4 are reported in Tables A.2 and A.3, respectively. They are consistent with the results in Table 3 in the main text with the threshold as 0.8.

A.3 Other Cutoffs

RateBud provides letter grades for the credibility of reviews of products, with A meaning excellent, B meaning very good, C meaning good, D meaning fair, E meaning poor, and F meaning caution. In the main text, we report the results with C, D, E and F as classified fake review products. Here, we rerun the product-level analyses with D, E and F as classified fake review products. The results are shown in Table A.4, which are largely consistent with the results in Table 4.

A.4 Normalizing Data

Another concern is that the results in Table 4 are driven by the fact that we normalize the metrics by dividing the total number of reviews. To mitigate this concern, in Figure A.1, we plot the total number of mentions of all keywords against the total number of reviews. It is clear in Figure A.1 that the majority of products with a higher share of fake reviews are above the regression line of products with a lower share of fake reviews. Additionally, the regression line for products with a higher share of fake reviews is steeper than and is above the regression line for products with a lower share of fake reviews. This result shows that no matter what the total number of reviews is, AI summarization systematically takes more reviews into sentiment analysis from the reviews of fake review products.

A further concern is that Amazon may only select a fixed number of reviews to extract keywords from. However, it is evident from Figure A.1 that the number of reviews that Amazon uses to extract keywords and analyze sentiments are steadily increasing with the total number of reviews without reaching an upper bound.

Appendix B Details on the Decomposition

B.1 Proof of Lemma 1

We introduce a dummy D_j . $D_j = 1$ if $j \in J_1$ and $D_j = 0$ if $j \in J_2$. Then, for an arbitrary variable Z_i , we have:

$$\begin{aligned}\mathbb{E}[Z_j D_j] &= p(D_j = 1) \mathbb{E}[Z_j | D_j = 1] \\ \mathbb{E}[Z_j] &= p(D_j = 1) \mathbb{E}[Z_j | D_j = 1] + p(D_j = 0) \mathbb{E}[Z_j | D_j = 0] \\ \mathbb{C}(Z_j, D_j) &= \mathbb{E}[Z_j D_j] - \mathbb{E}[Z_j] \mathbb{E}[D_j] = p(D_j = 1) \mathbb{E}[Z_j | D_j = 1] \\ &\quad - p(D_j = 1)[p(D_j = 1) \mathbb{E}[Z_j | D_j = 1] + p(D_j = 0) \mathbb{E}[Z_j | D_j = 0]] \\ &= p(D_j = 1)p(D_j = 0)[\mathbb{E}[Z_j | D_j = 1] - \mathbb{E}[Z_j | D_j = 0]]\end{aligned}$$

Therefore,

$$\mathbb{E}[Z_j | D_j = 1] - \mathbb{E}[Z_j | D_j = 0] = \frac{\mathbb{C}(Z_j, D_j)}{p(D_j = 1)p(D_j = 0)}$$

This means that the

According to Cauchy–Schwarz inequality,

$$|\mathbb{E}[Z_j | D_j = 1] - \mathbb{E}[Z_j | D_j = 0]| \leq \frac{\sqrt{\mathbb{V}(Z_j)\mathbb{V}(D_j)}}{p(D_j = 1)p(D_j = 0)} = \sqrt{\frac{\mathbb{V}(Z_j)}{p(D_j = 1)p(D_j = 0)}}$$

B.2 Robustness Check in a Worse Case

Table A.5 presents an even more conservative decomposition of $\mathbb{E}[w_j | j \in J_1] - \mathbb{E}[w_j | j \in J_2]$ with J_1 denoting non-Amazon products and J_2 denoting Amazon-sold products. Specifically, we replace the estimated baseline heterogeneity term B with the upper endpoint of its 95% confidence interval in the subsample of products priced higher than \$250, thereby imposing a more stringent standard against our interpretation. Even under this conservative specification, the smallest f_{WRMS} required to fully explain away the overrepresentation result would still need to be as high as 37.92%, which is implausibly large in our setting.

B.3 Details on Estimation of $\mathbb{V}(\delta_j)$

We first estimate $\delta_j \equiv \mathbb{E}[W_{ij} | F_{ij} = 1, j] - \mathbb{E}[W_{ij} | F_{ij} = 0, j]$. To approximate W_{ij} , we use the method explained in Section 6.1.2 to match the list of extracted keywords and the reviews. Then we can calculate average of W_{ij} of fake reviews and authentic reviews for each product. Finally, we take the difference between them and get $\hat{\delta}_j$.

Data source 2 is only a subsample of all reviews, which may bias the estimation of $\mathbb{V}(\delta_j)$. We use two empirical strategies to deal with this problem. First, we only select products with at least five authentic reviews and five fake reviews. Second, if we assume $\hat{\delta}_j$ is unbiased, there is a clean

way to debias the estimation of $\mathbb{V}(\delta_j)$ based on the law of total variance in the following equation, where $n_{1,j}$ and $n_{0,j}$ denote the number of fake and authentic reviews of product j respectively.

$$\begin{aligned}\mathbb{V}(\hat{\delta}_j) &= \mathbb{E}[\mathbb{V}(\hat{\delta}_j | j)] + \mathbb{V}(\mathbb{E}[\hat{\delta}_j | j]) = \mathbb{V}(\delta_j) + \mathbb{E} \left[\mathbb{V} \left(\frac{1}{n_{1,j}} \sum_{i:F_{ij}=1} W_{ij} - \frac{1}{n_{0,j}} \sum_{i:F_{ij}=0} W_{ij} \mid j \right) \right] \\ &= \mathbb{V}(\delta_j) + \mathbb{E} \left[\frac{\sigma_{F,j}^2}{n_{1,j}} + \frac{\sigma_{A,j}^2}{n_{0,j}} \right]\end{aligned}$$

Appendix C Robustness Checks for Distortion in Product Quality Information

C.1 Sentiment Analysis with Specialized BERT

In the main text, we use the famous general-purpose model OpenAI GPT-4.1 to extract the summary paragraphs to sentiment measured on the scale of ratings. Here, we use a specialized BERT model finetuned to predict ratings of reviews, named `bert-base-multilingual-uncased-sentiment`,³⁰ to convert summary paragraphs to sentiments measured on the scale of ratings. The results are shown in Table A.6.

C.2 Share of Time in Buy Box

A tricky point here is that products sold by Amazon when we collected the data may not always be sold by Amazon. On Amazon, different sellers of a product share the same product page and compete to be the featured offer prominently displayed in the Buy Box.³³ Amazon pools different sellers of one product together. To deal with this concern, we collect the history of the featured offer in Buy Box from Keepa. We confirm that during 80% of the time, the Buy Boxes of Amazon-sold products in our data were occupied by Amazon, while during only 6% of the time, the Buy Boxes of the other products were occupied by Amazon. This history is only available for a subset of products, so we still use Amazon-sold products as the proxy in the main text.

We present the results with the share of time during which the Buy Box of each product is occupied by Amazon as a proxy for how unlikely the products are to have fake reviews. We replace Amazon-sold indicator with the continuous Amazon share variable. The result is presented in Table A.7. The result is largely consistent with that shown in the main text.

C.3 Other Cutoffs

RateBud provides letter grades for the credibility of reviews of products, with A meaning excellent, B meaning very good, C meaning good, D meaning fair, E meaning poor, and F meaning caution. In the main text, we report the results with C, D, E and F as classified fake review products. Here, we rerun the analyses with D, E and F as classified fake review products. However, products

with grades D, E and F are scarce in our data, because we only collected products shown on the first several pages of the search results.

We compare classified fake review products with a stricter cutoff (C is not fake) and other non-Amazon products. The results are presented in Table A.8. Results are still significant and consistent with what is shown in the main text, despite the smaller size of classified fake review products.

Appendix D Appendix Figures and Tables

Table A.1: Review-Level Analysis with paraphrase-mpnet-base-v2

	Fake Reviews	Authentic Reviews	Authentic 5-Star Reviews
Panel A: Textual Features of Fake Review VS Authentic Reviews			
Cosine Similarity	0.5767	0.4045 ($p < 0.001$)	0.5077 ($p < 0.001$)
Panel B: Testing Hypothesis 1 by Comparing # Extracted Keywords Mentioned by Fake Reviews VS Authentic Reviews			
# Keywords Extracted by Amazon	0.4046	0.2357 ($p < 0.001$)	0.3344 ($p < 0.001$)
# Keywords Extracted by BERTopic	0.7302	0.5292 ($p < 0.001$)	0.5538 ($p < 0.001$)
# Keywords Extracted by Top2Vec	0.1640	0.0854 ($p < 0.001$)	0.1084 ($p < 0.001$)

Table A.2: Review-Level Analysis with the Threshold as 0.6 in Semantic Matching

	Fake Reviews	Authentic Reviews	Authentic 5-Star Reviews
# Keywords Extracted by Amazon	0.5083	0.2956 ($p < 0.001$)	0.4204 ($p < 0.001$)
# Keywords Extracted by BERTopic	0.7278	0.5374 ($p < 0.001$)	0.5270 ($p < 0.001$)
# Keywords Extracted by Top2Vec	0.3367	0.1992 ($p < 0.001$)	0.2413 ($p < 0.001$)

Table A.3: Review-Level Analysis with the Threshold as 0.4 in Semantic Matching

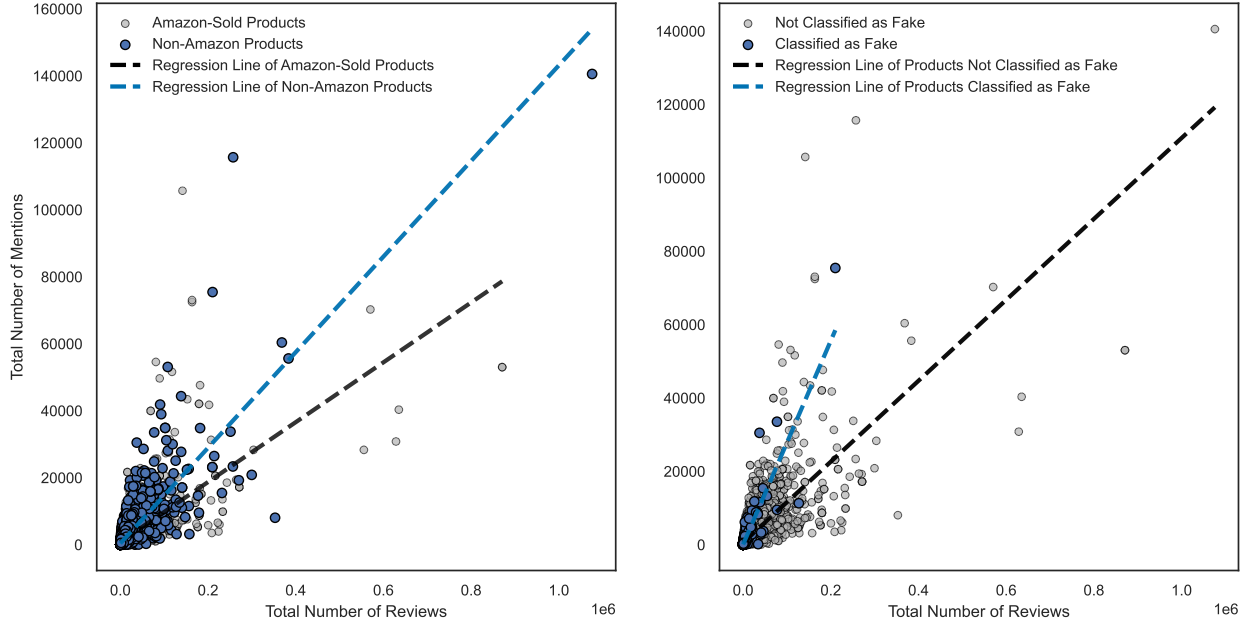
	Fake Reviews	Authentic Reviews	Authentic 5-Star Reviews
# Keywords Extracted by Amazon	1.2573	0.7514 ($p < 0.001$)	1.0481 ($p < 0.001$)
# Keywords Extracted by BERTopic	0.8504	0.6326 ($p < 0.001$)	0.6297 ($p < 0.001$)
# Keywords Extracted by Top2Vec	0.7929	0.5669 ($p < 0.001$)	0.6193 ($p < 0.001$)

Table A.4: Product-Level Evidence with C as Authentic

Sample	(1) All	(2) All	(3) Unbounded	(4) Unbounded	(5) All	(6) All
Amazon-Sold						-0.0986*** (0.00704)
Classified as Fake	0.485*** (0.0484)	0.434*** (0.0534)	0.529*** (0.0685)	0.478*** (0.0735)	0.253*** (0.0428)	0.236*** (0.0425)
# Keywords					0.0227*** (0.00207)	0.0239*** (0.00207)
Avg Rating					-1.349*** (0.0588)	-1.347*** (0.0584)
# Reviews					-1.71e-06*** (1.21e-07)	-1.56e-06*** (1.20e-07)
Variance of Ratings					-0.166*** (0.0128)	-0.167*** (0.0127)
Share of 5-Star Reviews					3.549*** (0.116)	3.534*** (0.116)
Constant	0.411*** (0.00324)	0.452*** (0.00417)	0.333*** (0.00639)	0.348*** (0.00751)	3.837*** (0.201)	3.859*** (0.200)
Search Term $\times$ Page No. FE	No	No	No	No	Yes	Yes
Category FE	No	No	No	No	Yes	Yes
Observations	14284	9658	2528	1916	13345	13323
R^2	0.007	0.007	0.023	0.022	0.503	0.511

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Figure A.1: Evidence for Overrepresentation of Fake Reviews without Normalization

Note: This figure plots the total number of mentions of all keywords, against the total number of reviews. The figure shows that for products with any total number of reviews, AI summarization algorithms take more reviews of fake review products into sentiment analysis and represent them in the sentiments of AI summaries.

Table A.5: Decomposition of $\mathbb{E}[w_j | j \in J_1] - \mathbb{E}[w_j | j \in J_2]$ with J_1 Denoting Non-Amazon Products And J_2 Denoting Amazon-Sold Products in a Worse Case

Quantity	Estimate
Estimated $\mathbb{E}[w_j j \in J_1] - \mathbb{E}[w_j j \in J_2]$	0.1259
- Upper Endpoint of 95% CI of B	0.0227
= Worse-Case $\hat{\delta}(\mathbb{E}[f_j j \in J_1] - \mathbb{E}[f_j j \in J_2]) + H$	0.1032
$\div$ Estimated $\sqrt{\mathbb{V}(\delta_j)/p(j \in J_1)/p(j \in J_2)}$	0.2722
= Smallest f_{WRMS} to rule out overrepresentation of fake reviews in the worst case	37.92%

Table A.6: Distortion in Product Quality Information with BERT

	(1) Paragraph	(2) Paragraph	(3) Paragraph	(4) Paragraph
Amazon-Sold	-0.0894*** (0.0114)	-0.0827*** (0.0114)		
Classified as Fake			0.173*** (0.0214)	0.160*** (0.0215)
Avg Rating	1.194*** (0.0202)	0.743*** (0.0939)	1.203*** (0.0203)	0.840*** (0.0953)
# Reviews		-9.28e-07*** (1.90e-07)		-1.14e-06*** (1.94e-07)
Variance of Ratings		-0.0420* (0.0203)		-0.0217 (0.0206)
Share of 5-Star Reviews		1.130*** (0.187)		0.992*** (0.189)
Constant	-1.724*** (0.0898)	-0.502 (0.322)	-1.799*** (0.0903)	-0.890** (0.327)
Search Term $\times$ Page No. FE	Yes	Yes	Yes	Yes
Category FE	Yes	Yes	Yes	Yes
Observations	13515	13515	13382	13382
R^2	0.433	0.436	0.434	0.437

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A.7: Distortion in Product Quality Information with Share of Time in Buy Box

	(1) Keyword	(2) Keyword	(3) Paragraph	(4) Paragraph
Amazon Share	-0.0725*** (0.00576)	-0.0703*** (0.00577)	-0.127*** (0.0107)	-0.121*** (0.0107)
Avg Rating	0.701*** (0.00877)	0.803*** (0.0407)	1.181*** (0.0163)	0.985*** (0.0754)
# Reviews		-1.77e-07* (8.02e-08)		-5.68e-07*** (1.49e-07)
Variance of Ratings		0.0588*** (0.00875)		0.0440** (0.0162)
Share of 5-Star Reviews		0.0621 (0.0820)		0.891*** (0.152)
Constant	-2.404*** (0.0389)	-2.983*** (0.139)	-1.039*** (0.0721)	-0.893*** (0.257)
Search Term $\times$ Page No. FE	Yes	Yes	Yes	Yes
Category FE	Yes	Yes	Yes	Yes
Observations	12753	12753	12753	12753
R^2	0.561	0.564	0.537	0.541

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A.8: Distortion in Product Quality Information with C as Authentic

	(1)	(2)	(3)	(4)
	Keyword	Keyword	Paragraph	Paragraph
Classified as Fake	0.0657* (0.0273)	0.0736** (0.0272)	0.168*** (0.0506)	0.160** (0.0505)
Avg Rating	0.675*** (0.00769)	0.824*** (0.0376)	1.138*** (0.0143)	0.990*** (0.0696)
# Reviews		-4.79e-07*** (7.24e-08)		-1.19e-06*** (1.34e-07)
Variance of Ratings		0.0605*** (0.00821)		0.0386* (0.0152)
Share of 5-Star Reviews		-0.0635 (0.0747)		0.726*** (0.138)
Constant	-2.308*** (0.0342)	-3.003*** (0.129)	-0.888*** (0.0634)	-0.817*** (0.239)
Search Term $\times$ Page No. FE	Yes	Yes	Yes	Yes
Observations	14315	14315	14315	14315
R^2	0.491	0.496	0.465	0.471

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$